



OPEN Smart deep learning model for enhanced IoT intrusion detection

Faisal S. Alsubaei

Growing volumes and sensitivities of information in the growing IoT require strong cybersecurity measures to adequately counter increasingly sophisticated cyberattacks. Machine learning-based anomaly detection has the potential to be a viable solution through abnormal network traffic behavior identification that foretells intrusions. Existing approaches, however, are usually hampered by the inability to effectively counter the sophisticated and evolving nature of such threats, especially in preprocessing optimization and hyperparameter tuning, which typically adopt conventional machine learning and deep learning models. This paper addresses these limitations with large preprocessing steps followed by hyperparameter tuning of machine learning XGBoost and deep learning Sequential Neural Network (OSNN) algorithms through Grid Search for their best values to improve multiclass intrusion detection across varied datasets. These deep models were then augmented with a variety of various filters, kernels, activation functions, and regularization techniques in an attempt to boost them in detecting complex, multiclass intrusion patterns. The proposed system was tested comprehensively on three challenging datasets: NSL-KDD, UNSW-NB15, and CICIDS2017. The optimized XGBoost model worked exceptionally well on the NSL-KDD dataset with very high accuracy (99.93%), F1-score (99.84%), MCC (99.86%), and a very low FPR (0.0004). The optimized SNN model also performed well on the NSL-KDD dataset with an accuracy of 99.0% and an AUC of 1.00. Also, the OSNN model performed very well on UNSW-NB15 dataset with an accuracy of 96.80% and a loss of 0.0777, as well as on the CICIDS-2017 dataset with an accuracy of 99.53% and a loss of 0.0236. This superb performance of the OSNN model can be explained by the careful optimization of hyperparameters like strong activation functions (ReLU, GeLU, LeakyReLU), learning rates, dropout rates, and regularization techniques that enable it to learn intricate intrusion patterns efficiently using various datasets. These results highlight the potential of our proposed method to enhance intrusion detection, system integrity, fraud prevention, and ultimately optimize overall network performance.

Keywords Internet of things (IoT), Cyber security, Intrusion detection, Intrusion detection systems (IDS), Optimized XGBoost, Optimized sequential neural network

Cybersecurity has become a mainstream keystone in today's globally interconnected world extending to all corners of society, from social and political structures to economics. Due to the constantly changing threat environment of cyberattacks, security researchers have developed a massive range of network intrusion detection (NID) mechanisms¹. Intrusion detection systems (IDS) silently keep an eye on network traffic for policy infractions or abnormal behavior, which are dynamic and adaptive rather than static rule-based firewalls. As our reliance on cyberspace and cloud computing grows exponentially, the need for robust cybersecurity has never been more necessary. From our daily lives to the most personal details, we permit social media and other internet portals to have, the safeguarding of our online lives from invasion is a complete necessity.

In our hyper-connected world, over 70% of our population navigates the digital cosmos on a sheer count of platforms, the internet offers huge potential as well as huge danger. While individuals and businesses bask in free information exchange without a glitch, cybercriminals utilize the same conduits, their exploits becoming even more sophisticated and complex to attack data and choke critical services. In an effort to address such shifting threats, researchers have increasingly turned to the use of machine learning (ML) and deep learning (DL) algorithms in designing sophisticated intrusion detection systems. The intricate nature of Internet of Things (IoT) networks and the explosive growth in data that they generate are significant limitations to traditional anomaly detection mechanisms.

Detection of known attacks is not enough to counter the dynamic attack methods of cybercriminals. This paper presents a novel machine learning-based anomaly detection algorithm for IoT systems. Our method is centered on detecting varied anomalies, i.e., in network traffic, in the extensive and dynamic IoT environment.

Department of Cybersecurity, College of Computer Science and Engineering, University of Jeddah, Jeddah, Saudi Arabia. email: fsalsubaei@uj.edu.sa

With the inclusion of a smart contextual understanding, it is able to detect normal and aberrant behavior with precision and subtlety, reducing false positives and maximizing detection ratios. Network communication, more critical in every industry, is now essentially a focal point for attack and peril. The most effective action against such a menace is marking network data as normal or suspicious. This method, which is known as “irregularity detection,” concentrates on the discovery of unexpected incidents in network traffic. Important departure from standard operation is defined as abnormal and activates additional analysis². Increased internet traffic tremendously amplified the threat of contemporary networks to all sorts of cyber-attacks. Amplified risk makes these networks susceptible to a wide array of malicious attacks such as data breach, disruption of service, and financial loss. For the sake of being able to effectively counter these threats, the creation of robust and effective Intrusion Detection Systems (IDS) is important in order to achieve this purpose. IDS are essential in maintaining the integrity of the network by pre-empting, detecting, and countering malicious activity.

The IoT IDS are designed to automatically detect and respond in real time against potential intrusions or security breaches. IoT IDS operate by monitoring network traffic, device behavior, and communication patterns in order to find anomalies, suspicious activity, or known attack signatures. Common techniques employed in IoT intrusion detection include:

Signature-based detection: This detection identifies potential intrusions by scanning network traffic or device behavior against a database of known attack signatures or patterns. Upon discovering a match, it raises an alarm.

Anomaly-based detection: This detection builds a profile of normal behavior for IoT devices and networks. It then identifies anomalies from the profile, in which abnormal or unusual behavior can signal a potential intrusion.

Behavioral detection: With this approach, a single IoT device's regular behavior is monitored. By establishing baseline patterns of usage for every device, the system will report deviations that indicate possible compromise or abuse. Machine learning algorithms are particularly important here in that they can be designed to monitor IoT information for search and identification of typical patterns indicative of malicious usage. These algorithms can learn from historical patterns and adapt to emerging threats.

Anomaly detection ability is crucial across various industries because it enables one to detect abnormal patterns, events, or behavior that are not normal. Anomaly detection feature is applicable in most industries such as finance, cybersecurity, health, and manufacturing, among others. The burgeoning IoT universe is defined by an avalanche in the volume and sensitivity of data, generating a critical necessity for sophisticated cybersecurity measures adept at protecting against increasingly sophisticated cyber threats. While anomaly detection via machine learning possesses great promise for detecting suspicious network behavior indicating intrusion, current methods often fail to measure up to the changing sophistication and dynamism of these threats. One of the major challenges is optimizing data preprocessing and model hyperparameter tuning, both domains where existing solutions often rely on conventional machine learning and deep learning architectures.

This work tackles these challenges by using comprehensive preprocessing techniques and rigorously optimizing the hyperparameters of the XGBoost machine learning algorithm and an OSNN deep learning model through Grid Search. The aim is to realize better multiclass intrusion detection performance on diverse datasets. Further, the research investigates the effect of including varied filters, kernels, activation functions, and regularization methods in the optimized deep learning models so as to maximize their ability to identify intricate, multiclass intrusion patterns. The proposed new technique has as a goal the detection of varied anomalies, and specifically anomalous network traffic, in the complex and large-scale IoT structure. Using this approach, we can achieve high detection rates and low false positives when distinguishing normal and abnormal traffic patterns. This enhances the security of sensitive information as well as IoT devices, hence optimizing overall security in IoT scenarios.

To overcome these challenges, this research presents several major innovations and contributions to improve intrusion detection performance:

- **Improved preprocessing:** We begin with a stringent preprocessing step to ensure optimal possible data preparation for subsequent analysis. This comprises several critical steps such as data normalization, balancing techniques (e.g., sub-sampling), categorical variable encoding, and feature extraction. This comprehensive data preparation is the foundation upon which the model can effectively derive meaningful inferences from the network traffic.
- **XGBoost model fine-tuning:** We perform the XGBoost model hyperparameter tuning cautiously through intensive grid search in pursuit of obtaining maximal detection precision. This tuning generates an ideal model to utilize in identifying very light intrusion patterns on the network traffic.
- **Optimized sequential neural network:** Continuing in the effort to enhance the flexibility of models, we use a custom sequential neural network model structure. The network is optimized for different hyperparameters, such as filter sizes, kernel sizes, pooling methods, dense layer sizes, and learning rates. We also test different activation functions, including ReLU, GeLU, and LeakyReLU, to achieve maximum model performance. Moreover, dropout layers are placed strategically to promote generalization and lower computational costs. The SNN model is optimized using well-tuned convolutional blocks, flattening blocks, and dense blocks with associated parameters to efficiently extract the unique signatures of different attacks.
- **Cross-validation and regularization:** To combat the weaknesses of overfitting, we employ two robust countermeasures: 5-fold cross-validation and L2 regularization. These techniques efficiently adapt to potential biases induced by class imbalance datasets, implementing a generalized and sound intrusion detection system.

Related work

Earlier studies³ have analyzed hybrid methods, including the integration of Random Forests (RF), decision trees, (DTs) and Support Vector Machines (SVM), for network intrusion detection. While effective, these techniques

are limited by lower detection rates and increased false alarm rates, which limit their capacity to fully prevent intrusions. Following this, later research⁴ tested a successful ML ensemble method for intrusion detection systems with particular attention to improving detection quality. The study highlighted the importance of adequate parameter fine-tuning in ML models and the use of strong pre-processing methods to enhance the detection of various types of intrusions, most importantly the rare attacks. Adeel et al.⁵ explained the use of unlabeled data with a combination of the Naive Bayes (NB) classifier and the Expectation-Maximization (EM) algorithm. Existing research has shown good usage of NB and decision tree algorithms in Network Intrusion Detection Systems (NIDS) with excellent accuracy under varying network scenarios⁵. One shared disadvantage of these methods is reliance on individual classifiers for individual attack classes, compared to aggregated class usage. Class aggregation gives probable reduced reliance on training data while identifying attacks, enabling easier model construction and better acceleration in testing.

Therefore, the combination of Feature Selection (FS) and classifier methods can potentially be used to improve performance⁶. Our study repeatedly highlights the combination of classifiers with different FS methods. Although SVM and artificial neural network (ANN) models tend to show better performance than other classification models, their use can be limited by the extensive training time for large datasets. In order to overcome this constraint, we suggest combining several feature selection (FS) methods with ANN and SVM classifiers in a way that facilitates precise attack detection with reduced computational overhead.

Machine learning and deep learning based literature background

For increased detection rates at the expense of reliability, writers⁷, proposed a new hybrid model incorporating machine learning and deep learning algorithms. This process efficiently preprocessed data through feature selection with the help of XGBoost and balancing of data using Synthetic Minority Over-sampling Technique (SMOTE). Later in their work, Ismail et al.⁷, examined a way of optimizing network parameters by the utilization of convolutional neural networks (CNNs) and Gated Recurrent Units (GRUs) to achieve intrusion detection. They investigated different combinations of CNN and GRU layers and performed simulations using the CICIDS-2017 benchmark dataset, measuring performance based on measures like recall, precision, false positive rate (FPR), true positive rate (TPR), and other appropriate metrics.

Noting that various models are capable of deriving information about data features from multiple angles, the authors in⁸ propose a new hybrid IDS that combines the benefits of both RF and Autoencoders (AEs). The proposed hybrid model works in two stages. During the initial stage, the RF classifier's probability output is employed to establish if a sample is an attack or not. This probability output can also be used to determine potential unknown attacks. To further minimize the false positive rate, a second Autoencoder (AE) is incorporated in the second phase.

Contemporary benchmark intrusion detection datasets, while having a varied mix of attacks, tend to be class-imbalanced with a much larger number of normal traffic compared to attack traffic^{9,10}. This class imbalance heavily detracts from the efficiency of NIDS as it makes it difficult to properly identify certain kinds of attacks. Whereas class imbalance hurts the performance of NIDS, this serious issue has not yet been given its due attention in recent NIDS studies^{11,12}. This work introduces a new hybrid intrusion detection classification model based on ML and DL methods that aims to increase detection rates and accuracy.

Tavallae et al.¹³ created the NSL-KDD dataset by refining the KDD99 dataset to primarily eliminate redundancy. This dataset has been extensively explored using various ML and DL algorithms. Khraisat et al.¹⁴ and Ever et al.¹⁵ demonstrated the effectiveness of numerous ML techniques on the NSL-KDD dataset, achieving high accuracy in intrusion detection. Zhang et al.¹⁶ implemented a DL model employing Autoencoders on the NSL-KDD dataset. However, it's crucial to acknowledge that these datasets are nearly two decades old. Apart from their inherent limitations, they may not adequately represent contemporary attack scenarios, given the rapidly evolving threat landscape.

Sperotto et al.¹⁷ introduced an IP flow-based intrusion detection dataset that categorizes attacks and defense mechanisms by identifying scans, worms, botnets, and DoS attacks. However, this dataset lacks representations of normal user behavior. Shiravi et al.²⁵ addressed this limitation by incorporating network traffic from typical users. Nevertheless, it lacks communication with external servers, a crucial aspect of real-world attack scenarios, as many attacks originate from external sources. Consequently, an IDS trained solely on internal server data may fail to accurately replicate and detect real-world attacks.

Smamarwar et al.¹⁸ developed an ensemble learning framework for Android malware detection (OEL-AMD), achieving high accuracy (96.95% binary, 83.49% category) by combining feature engineering, Binary Grey Wolf Optimizer (BGWO)-based selection, and hyperparameter tuning. The study¹⁹ comprehensively reviews malware detection approaches, categorized by feature selection, machine learning, and deep learning techniques, identifying research gaps and future directions for developing more effective and resilient malware detection frameworks. The researchers²⁰ proposes a novel hybrid feature selection framework combining Random Forest and Greedy Stepwise selection (RF-GreedySW) to optimize malware features, achieving high classification accuracy (up to 91.80%) on the CIC-InvesAndMal2019 dataset using ML models like RF, Decision Tree, and SVM-RBF. The researchers^{19–25} recently developed deep learning method for improving detection of malware in various fields for IoT based smart industrial, smart agriculture.

In-vehicle networks face increasing cyber threats, and existing intrusion detection systems struggle to effectively identify and mitigate these risks. The researchers²⁶ introduces HDL-IDS, a hybrid deep learning framework that leverages knowledge graphs, CNNs, and long-short term memory (LSTM) to accurately detect both known and unknown attacks in in-vehicle networks, achieving up to 99% accuracy with low false positives. HDL-IDS addresses key limitations of existing systems by incorporating semantic analysis, capturing both spatial and temporal dependencies in network traffic, and effectively detecting anomalies in non-periodic messages, paving the way for enhanced security in connected vehicles. The authors²⁷ proposes a hybrid deep learning

model combining CNN and Bi-LSTM, achieving high accuracy, precision, and low FPR in detecting network intrusions on NSL-KDD, and UNSW-NB15 datasets, outperforming existing methods.

Lo et al.²⁸ proposes HyDL-IDS, a hybrid deep learning model combining CNN and LSTM to accurately characterize in-vehicle network traffic and detect cyberattacks (DoS, fuzzy, spoofing) on CAN buses with highest detection accuracy and low false alarms, outperforming traditional machine learning models. The researchers²⁹ presents a comprehensive review of recent research on ML based IDS in Software-Defined Networking environments, analyzing existing works, discussing research challenges, and outlining promising future directions for effective and efficient intrusion detection in SDN.

Designing robust and efficient Intrusion Detection Systems (IDS) for IoT networks presents significant challenges, primarily due to the unpredictable and unknown nature of attacks, imbalanced datasets, and high dimensionality. Recent research³⁰ proposed a transformer-based IDS with multi-head self-attention for feature selection and SMOTE for addressing class imbalance, achieving 98.4% accuracy on CICIDS2017 and CIC-DDoS2019 datasets. Another study³¹ introduced a hybrid CNN-LSTM IDS, achieving 98% accuracy, but lacked detailed architectural explanations. A semi-supervised deep learning IDS³², utilizing a multi-scale residual temporal convolution module and attention mechanism achieved 99.69% accuracy, although its performance with massive traffic and unknown attacks remains unclear.

Datasets and their evaluations-based background

The researchers³³ embraced the power of transfer learning, a technique that repurposes knowledge gained from one task to another. They cleverly transformed network traffic data into images using chunk-based processing, enabling the utilization of a powerful CNN. Additionally, ensemble learning, which combines the strengths of multiple models, bolstered performance. This hybrid approach outperformed existing models on datasets like CICIDS2017 and car hacking data, achieving a superior F1 score. Another research team³⁴ turned to vision transformers, a deep learning architecture revolutionizing image recognition. They developed a pre-processing technique that converted network traffic features into four-channel images, suitable for feeding into a pre-trained ResNet-50 model. This sophisticated strategy leveraged the pre-trained model's knowledge to label attacks effectively, showing improved accuracy on BOUN DDoS and UNSW-NB15 datasets. Imbalanced data, in which malicious instances greatly outbalance benign ones, is a common difficulty in IDS training^{35–38}. Foreseeing this, Yuelei Xiao et al.³⁶ proposed a reduced residual network (ResNet) with a Parametric Rectified Linear Unit (PReLU) activation function. This adaptation functioned effectively in dealing with data skewness and produced greater accuracy and recall than the conventional IDS models. Martin Kody et al.³⁹ also brought out the effectiveness of CNN over sequential deep learning models for the use in IDS. They utilized two different CNN architectures, EfficientNet and ResNet-50, on the TON IoT Dataset to measure their efficiency. Both models performed better than previous IDS structures in recall, testifying to the capability of CNNs in handling complex network traffic patterns. ResNet is superior even outside biased data since another article⁴⁰ depicts. The authors employed ResNet without data conversion in this scenario and achieved an astonishing 99.9% accuracy in detection of DDoS and DoS attacks, further solidifying its reputation as a reliable intrusion detection instrument. These early attempts reveal the rich and promising areas developing in IDS for IoT. Each model discusses unique challenges, demonstrating the extreme potential of deep learning and its related methods. As more experiments are conducted, we can look forward to continuously sophisticated and more responsive systems that will safeguard an ever-expanding universe of interlinked devices.

Jiarui Song et al. (2020)⁴¹ designed a novel neural network with the combined strength of attention mechanisms, LSTMs, and ResNets. They work on extracting distinctive patterns from network traffic data that are then improved with a feature selection mechanism that they develop on Light Gradient Boosting Machines (GBM), resulting in powering a significantly efficient IDS model outperforming current systems. Building on the powerful transformational abilities of vision transformers, yet another research work (2023)³⁸ introduces an innovative IDS model. This hybrid structure marries ResNet-50 and CNNs, proving effective in converting numerical datasets into image formats that can be processed by transformers. Tested on established benchmarks such as UNSW-NB15, NSL-KDD, and CICIDS2017, the model exhibits high accuracy levels over 90%, reflecting the massive potential of vision transformers to enhance IDS performance. ResNet's ability to generalize extends beyond data imbalance, as shown by Tala et al. (2022)⁴². Their ResNet-50-based IDS targets the smart grid domain and utilizes the "Deep Insight" approach to transform numerical data into images for processing. Although superior in denial-of-service attacks detection, the model's ability for other types of attacks is not explored. Kumar et al. (2022)⁴³ propose a new IDS that couples a recurrent kernel CNN with a Modified Monarch Butterfly Optimization (MMBO) algorithm. Leverage the strength of min-max pre-processing coupled with MMBO-based feature extraction, their model attains astounding accuracy (99.5% on CICIDS-2017 and 99.96% on N-BaIoT) while illustrating the potential of using Modified Monarch Butterfly Optimization (MMBO) for improving the performance of deep learning classifiers. Another research in 2021⁴⁴ touted the strength of Bi-directional LSTMs for IDS development. Their framework achieved outstanding accuracy well over 99% in both UNSW and BOT-IOT datasets, which further affirmed the strength of LSTMs for intrusion detection and further solidified its role as a force to be reckoned with in this field. These papers offer only a glimpse of the rich IDS research landscape, highlighting the varied arsenal of methods used to counter increasingly sophisticated cyber threats. Through collaboration and ongoing innovation, researchers and developers can make certain that the strong and wise sentinels that protect our networked world are prepared to meet the challenges of the future.

The datasets utilized in this study, sourced from the Indus experimental IoT environment^{45,46}, encompass a comprehensive range of potential attack scenarios and provide rich sample sizes. In the contemporary era of digital technologies, network systems facilitate the transmission of diverse and heterogeneous data across distributed environments. However, this interconnectedness has also led to a significant increase in network security threats, driven by the rise of cybercrime, the advancement of communication technologies, and

the expansion of attack surfaces. Therefore, mitigating and ideally preventing these threats is of paramount importance.

Researchers have recently employed innovative methods to enhance both the IoT and Intrusion Detection Systems. The growing number of IoT networks increased cybersecurity threats, calling for efficient IDS; however, the IDS in use today utilize computationally intensive deep learning and fail to detect complex IoT-targeted attacks as a result of their poor capacity to recognize temporal dependencies and their non-transparency. To address these challenges, researchers⁴⁷ employed DistillGuard, a new IDS framework, a Transformer-based teacher model with hybrid attention and selective knowledge distillation to a lightweight student, demonstrating better accuracy, efficiency, and explainability by gradient-based analysis.

The methods utilized by⁴⁸ faced with escalating cybersecurity threats and imperative to safeguard digital assets, businesses are increasingly relying on information systems for decision-making. Threat hunting offers a proactive, data-driven strategy for real-time cyber threat detection and response, yet it encounters challenges like manual analysis, non-Gaussian data, high false positives, and a lack of model interpretability. They⁴⁸ introduces DeepSecure, a novel IT artifact developed using computational design science, which employs a dynamic vector quantized variational autoencoder for latent pattern extraction and a multiscale hierarchical attention Bi-GRU for threat hunting, further providing visualization for model interpretation. Evaluated on CIC-IDS2018 datasets, DeepSecure demonstrates efficient identification of threat types and offers improved interpretability for quicker organizational response to security incidents.

The author⁴⁹ used Software-Defined Networking (SDN) with the IoT SDN-IoT offers centralized management but poses great security threats since a lot of data is involved. To tackle these challenges⁴⁹, introduced a novel Deep-Learning empowered Threat Hunting Framework (DLTHF), that utilizes an unsupervised feature extraction module with LSTMCSAE, and a threat detection system (TDS) based on MhSaBiGRNN with a self-learned weighting technique for every time step, proving its efficacy on novel SD-IoT datasets through a suggested deployment architecture.

The growing population of networked smart devices managing sensitive information exposes the IoT to cyberattacks, requiring efficient (IDS, which tend to have high False Positive Rates (FPR). The author⁵⁰ suggested a cyber-attack detection methodology for IoT that employs data preprocessing, optimized feature selection using an improved Binary Pigeon Inspired Optimizer, and classification with monolithic (kNN, SVM, DT) and ensemble (Random Forest) models with the best performance on the UNSW-NB15 dataset.

Conventional intrusion detection in cyber-physical systems and IoT generally employs point anomaly detection, making inefficient use of available multivariate time series data. The author⁵¹ presents MultiCIDS, an innovative method detecting collective intrusions by considering monitoring data as multivariate time series, utilizing a hybrid approach of a semi-supervised autoencoder for per-point scoring and a sliding window algorithm, which yields competitive performance, zero processing time, and heavily decreased false positives on various benchmark datasets. The researchers⁵² proposed a new Transformer for IDS in IoT, that substitutes Multi-Layer Perceptron (MLP) layers with Kolmogorov–Arnold Networks (KANs) layers with learnable univariate functions for more efficient representations.

With this meticulously designed framework, we aim to provide IoT networks with a strong defense against the ever-evolving nature of cyber threats. Figures 1 and 2 is a schematic diagram of this general approach. Figure 1 shows the proposed model to detect the intrusion by tuning the hyperparameters of Sequential neural network and XGBoost:

Figure 2 depicts a typical machine learning cycle for intrusion detection system design. The cycle commences with the utilization of renowned datasets, namely NSL-KDD, UNSW-NB15, and ICIDS2017. Preprocessing is extensive, involving normalization and class imbalance handling methods applied to these datasets. Then features are extracted that are suitable from the preprocessed data. Training and test sets are created to assess the model. A critical step is the selection and tuning of suitable machine learning algorithms. This step centers on algorithms like XGBoost and OSNN, with hyperparameter tuning methods like randomized search utilized to determine optimal settings. The learned models are subsequently thoroughly tested with both validation and independent test sets. Lastly, the model performance is thoroughly evaluated under different scenarios, such as binary and multi-class intrusion detection tasks.

Materials and methods

Dataset description

This research involves the rigorous evaluation on three challenging datasets (CICIDS2017, NSL-KDD, and UNSW-NB15), and demonstrated the effectiveness of the proposed intrusion detection system. The first dataset NSL-KDD dataset is publicly available at (<https://data.mendeley.com/datasets/t5bfffjd28/1>) and (<https://www.kaggle.com/datasets/hassan06/nslkdd>) without any restrictions under license CC0 1.0 and used by⁵³. The dataset is also utilized in³⁶ is a crucial training ground for intrusion detection systems. The attacks are further divided into subtypes, resulting in a total of 23 attack subtypes. It offers a realistic picture of network traffic, including both normal and malicious activities, with 41 detailed features like duration and attack types. These features are further categorized for focused analysis. Importantly, NSL-KDD addresses biases and redundancies compared to its predecessor, the KDD Cup 99 dataset^{54,55}. Two versions, original and reduced, cater to both large-scale and manageable research needs. Researchers have extensively employed NSL-KDD to evaluate various ML and DL techniques, as well as feature selection strategies, solidifying its role as a cornerstone in IDS development. In essence, NSL-KDD empowers researchers to refine and strengthen the digital sentinels protecting our interconnected world. The 41 features are divided into four types of categories shows in Table 1.

The NSL-KDD cup dataset contains four primary attack classes:

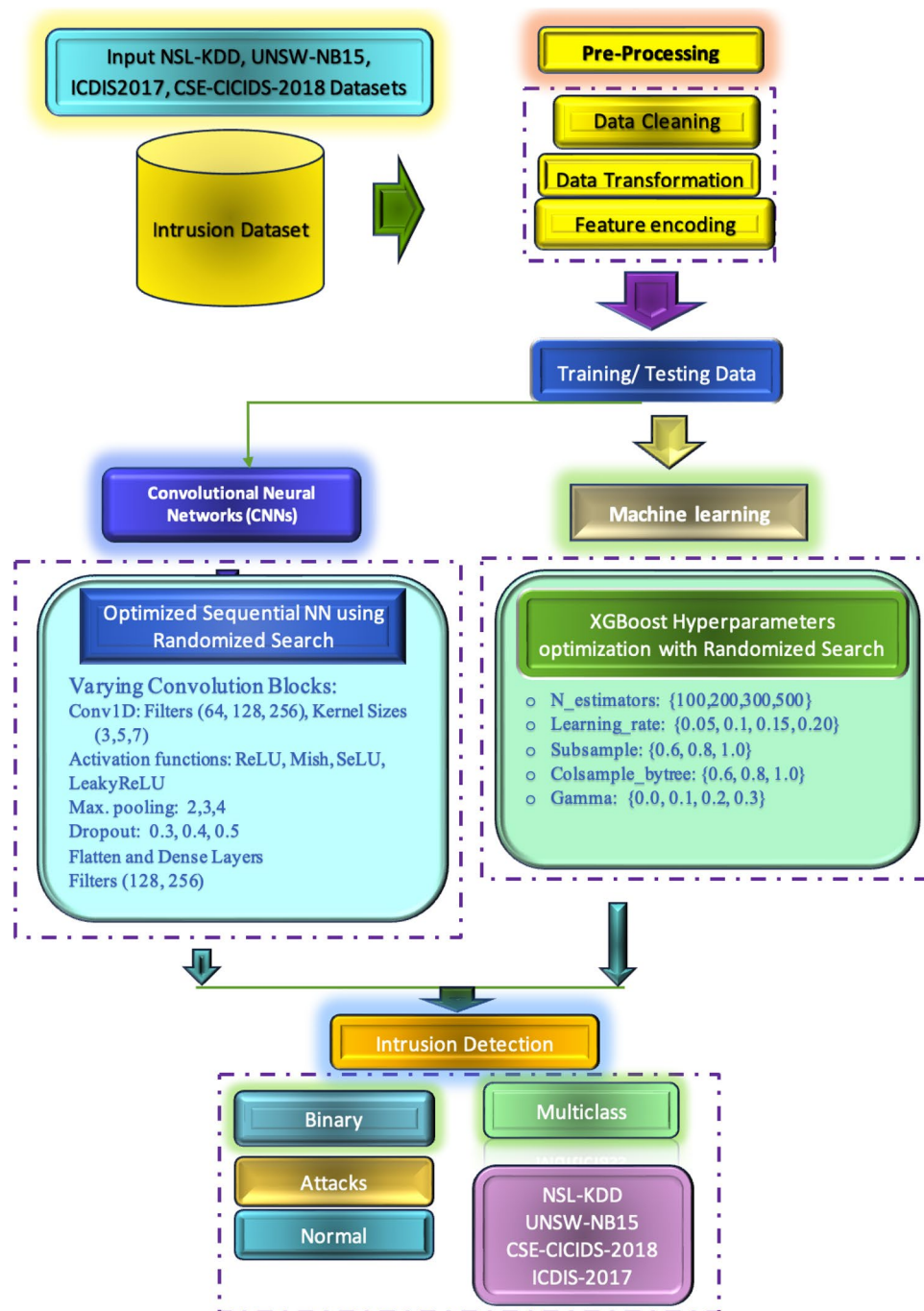


Fig. 1. Schematic diagram for intrusion detection.

1. Probing Attacks: These attacks serve as reconnaissance missions for malicious actors. By analyzing source bytes and connection duration, patterns indicative of probing attacks can be identified⁵⁶. Their goal is to uncover valuable information about target systems or networks, such as IP addresses, operating systems, and active services.
2. DoS Attacks: These attacks aim to overwhelm systems and hinder their ability to respond to legitimate requests.
3. R2L Attacks: These attacks seek to seize unauthorized local access to systems from remote locations. Vulnerabilities in operating systems, applications, or network infrastructure are often exploited to achieve this goal.
4. U2R Attacks: These attacks aim to escalate privileges, seeking to gain root or administrator access to systems.

This dataset consists of three files. Total record of KDD Train + 20% is 25,192. Other division of classes are mentioned in the Fig. 3 (a-c):

The Table 2 reflects the attack types and their classes.

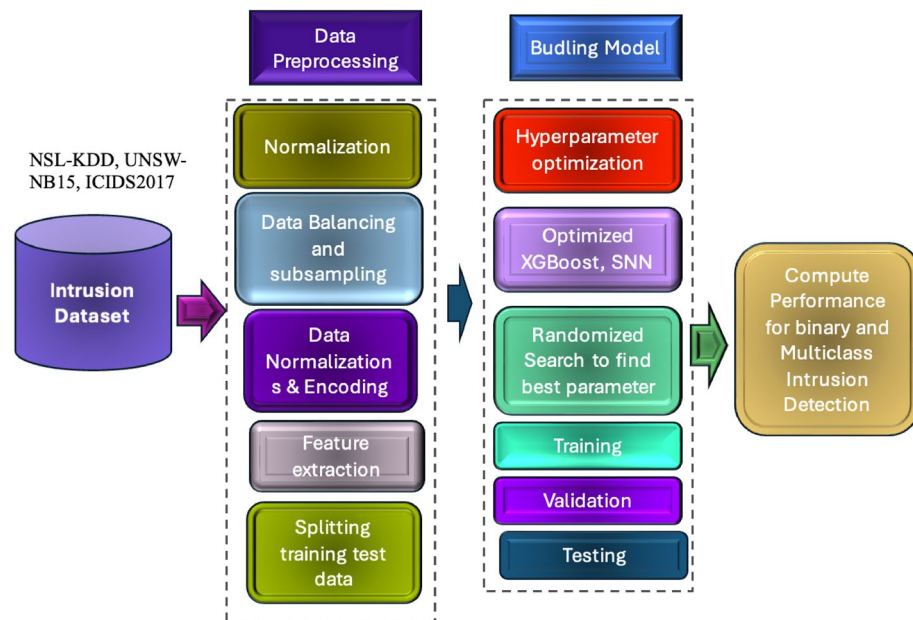


Fig. 2. Proposed framework.

Feature #	Feature type
1–9	Basic feature
10–22	Content related feature
23–31	Time related traffic feature
32–41	Host based traffic feature

Table 1. Types of features.

The proposed models are also verified on other publicly available datasets such as UNSW-NB15, and CICIDS2017.

Extreme boosting (XGBoost)

XGBoost, a powerful machine learning algorithm, is based on gradient boosting, sequentially training decision trees to improve predictions^{57–59}. It incorporates several improvements, including regularization for stability, efficient handling of sparse data, and parallel processing for speed. This versatility makes XGBoost a valuable tool for various applications, from fraud detection to medical diagnosis, excelling in both classification and regression tasks. XGBoost optimizes a loss function combined with regularization, iteratively building trees focused on reducing the loss. The final prediction is the sum of all tree predictions. Its high performance, efficiency, flexibility, and partial interpretability have established XGBoost as a preferred choice for solving complex prediction problems across diverse domains^{60–63}. XGBoost, being an ensemble learning technique, is an extension of the gradient boosting idea where a sequence of decision trees is trained sequentially. Each of the subsequent trees tries to correct the mistakes of the previous ones. XGBoost, being a high-performing machine learning technique, has some important additions to its credit that make it highly performant in various applications. They are regularization to avoid overfitting, optimized sparse data handling, and parallelization for better training speed. In addition, XGBoost has cross-validation built into it for stable model evaluation. These features combined make XGBoost a general-purpose tool that can be used across many fields, including fraud detection, customer churn prediction, medical diagnosis, and image classification^{64–66}.

Gradient boosting is divided into two primary components: optimization step and step direction. These components work together to refine the model's predictions as reflected in Eq. 1 to 9:

- But the XGBoost solve,

$$\frac{\partial S(y, f^{(m-1)}(x) + f_m(x))}{\partial f_m(x)} = 0 \quad (1)$$

For each data point x , directly adjust the step size. We have:

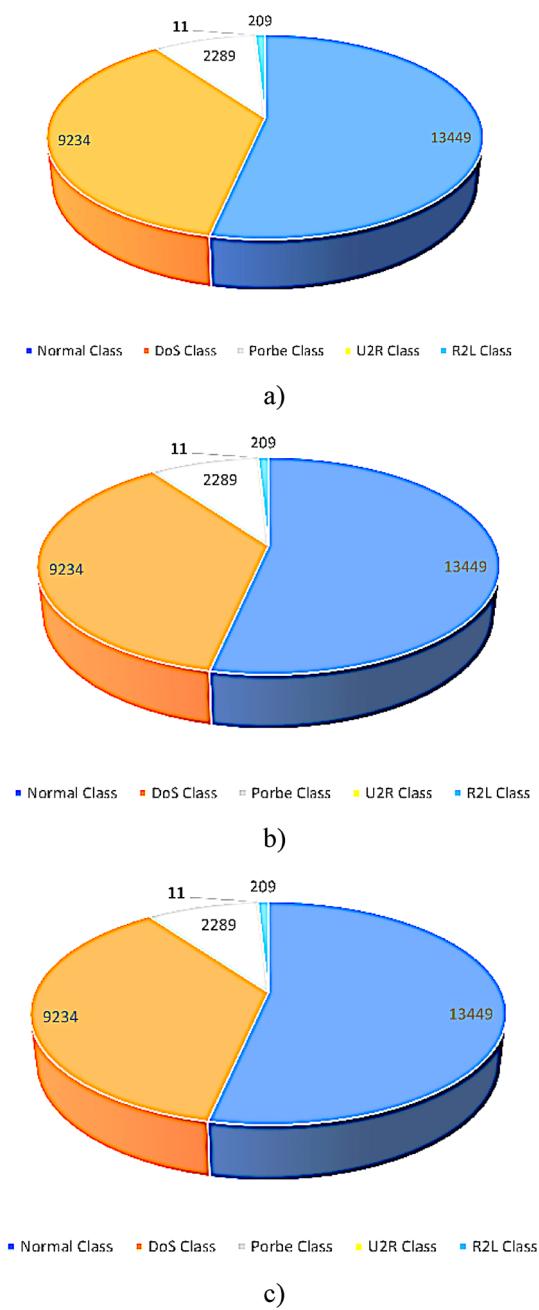


Fig. 3. Intrusion detection attack details in (a) KDD Train + 20, (b) KDD Train+, (c) KDD Train+.

Attack class	Types of attacks
Probe	Ipsweep, Satan, Mscan, Portsweep, Nmap, Saint (6)
DoS	Back, Land, Smurf, Processtable, Teardrop, Neptune, Apache2, Pod, Udpstorm, Worm
U2R	Ps Buffer_overflow, Rootkit, Sqlattack, Xterm, Loadmodule, Perl
R2L	Guess_Password, Warezmater, Ftp_write, Imap, Snmpgetattack, Phf, Multihop, Warezcclient, Named (16), Xlock, Xsnoop, Spy, Snmpguess, Sendmail, Httptunnel

Table 2. Attack classes and their Types.

Input:	
(a) Dataset consisting of Intrusion Detection data X and output labels Y . Each label $y \in$ Various classes of Intrusion type	
(b) Hyper-Parameters: $epochs_{total}$, $learning_rate$, n estimators, max_depth , $colsamples$, $sub-samples$, etc.	
Output: A directory containing the best model, plots, evaluation metrics, etc.	
1	Apply Preprocessing $dataset_{\{Missing, Normalization, Encoding\}}$
2	Prepare and create $dataset_{train}$ and $dataset_{test}$ using train-test_split method
3	Create XGBoost model object $Optimizer$, $Scheduler_{LR}$, and $Loss_{Function}$
4	Define parameter distributions for Randomized Search $param_grid = \{$ $"n_estimators": [100, 200, 300, 500],$ $"subsample": [0.6, 0.8, 1.0],$ $"Colsample_bytree": [0.6, 0.8, 1.0],$ $"gamma": [0, 0.1, 0.2, 0.3]$ $"learning_rate": [0.05, 0.1, 0.15, 0.2],$ $\}$
5	Optimize XGBoost Model: $random_search_XGBoost = RandomizedSearchCV(param_distributions=param_grid_XGBoost,$ $estimator=model_XGBoost, n_iter=50, cv=3, n_jobs=-1)$
6	Apply Early Stopping: $early_stopping_XGBoost = EarlyStopping(monitor='val_loss', patience=10,$ $restore_best_weights=True)$
7	Train and Evaluate XGBoost: $random_search_XGBoost.fit(X_train, y_train, epochs=100, validation_split=0.2,$ $callbacks=[early_stopping_XGBoost])$
8	Save the best XGBoost model.
9	Generate Evaluation metrics and Plots to show XGBoost performance.
10	Evaluation metrics and Plots to show performance

Algorithmic 1. Run complete experiment.

$$S(y, f^{(m-1)}(x) + f_m(x)) \quad (2)$$

$$\approx S(y, f^{(m-1)}(x)) + g_m(x) f_m(x) + \frac{1}{2} h_m(x) f_m(x)^2 \quad (3)$$

By employing a second-order Taylor expansion to approximate the loss function, where $g_m(x)$ is gradient and $h_m(x)$ is Hessian.

$$h_m(x) = \frac{\partial^2 S(Y, f(x))}{\partial f(x)^2}, \text{ here } f(x) = f^{(m-1)}(x) \quad (4)$$

- The loss function can be rewritten as follows:

$$S(f_m) \approx \sum_{i=1}^n \left[g_m(x_i) f_m(x_i) + \frac{1}{2} h_m(x_i) f_m(x_i)^2 \right] + const \quad (5)$$

$$\propto \sum_{j=1}^{P_m} \sum_{i \in R_{jm}} \left[g_m(x_i) K_{jm} + \frac{1}{2} h_m(x_i) K_{jm}^2 \right] \quad (6)$$

- In region j , lets G_{jm} represent sum of gradient and sum of Hessian is denoted by H_{jm} , then equation will be,

Input:	
(a) Dataset consisting of Intrusion Detection data X and output labels Y . Each label $y \in$ Various classes of Intrusion type	
(b) Hyper-Parameters: $epochs_{total}$, $learning_rate$, filter, kernel size, $weight_decay$, dropout rate, l2 regularization etc.	
Output: A directory containing the best model, plots, evaluation metrics, etc.	
1	Apply Preprocessing $dataset_{\{Missing, Normalization, Encoding\}}$
2	Prepare and create $dataset_{train}$ and $dataset_{test}$ using train-test_split method
3	Create Optimized SNN Model objects $Model$, $Optimizer$, $Scheduler_{LR}$, and $Loss_{Function}$
4	Define the OSNN model architecture with Convolution, pooling, dropout, Fatten, Dense with different kernel size, filters, regularization terms, and learning rates.
5	Define parameter distributions for Randomized Search $param_dist = \{$ 'filters': [32, 64, 128, 256], 'pool_size': [2, 4], 'kernel_size': [3, 5, 7], 'dropout_rate': [0.2, 0.3, 0.4, 0.5], 'dense_units': [50, 100, 150, 200], 'activation': ['relu', 'tanh', 'gelu', 'LeakyReLU', 'mesh'], 'learning_rate': [0.001, 0.0001, 0.00001] 'l2_regularization': [0.01, 0.001, 0.0001] }
6	Optimize OSNN Model: $random_search_OSNN = RandomizedSearchCV(param_distributions=param_grid_OSNN,$ $estimator=model_OSNN, n_iter=50, cv=3, n_jobs=-1)$
7	Apply Early Stopping: $early_stopping_OSNN = EarlyStopping(monitor='val_loss', patience=10,$ $restore_best_weights=True)$
8	Train and Evaluate OSNN: $random_search_OSNN.fit(X_train, y_train, epochs=100, validation_split=0.2,$ $callbacks=[early_stopping_OSNN])$
9	Save the best OSNN model.
10	Generate Evaluation metrics and Plots to show OSNN performance.
11	Evaluation metrics and Plots to show performance

Algorithmic 2. Run complete experiment.

$$S(f_m) \propto \sum_{j=1}^{P_m} \left[G_{jm} K_{jm} + \frac{1}{2} H_{jm} K_{jm}^2 \right] \quad (7)$$

- The optimal value can be calculated using the following equation.

understanding of model optimization procedures

$$K_{jm} = -\frac{G_{jm}}{H_{jm}}, \text{ where } j = 1, 2, \dots, P_m \quad (8)$$

- Substituting the optimal value into the loss function, we obtain:

$$S(f_m) \propto -\frac{1}{2} \sum_{j=1}^{P_m} \frac{G_{jm}^2}{H_{jm}} \quad (9)$$

The evaluation of the tree structure is performed using this function. A lower score signifies a more optimal tree structure⁹². The gain and loss were computed as utilized in⁶⁷.

Hyperparameters optimization

Hyperparameter tuning of XGBoost with Grid Search has a strong scientific basis in its exhaustive search of parameter space. Several research works^{68,69} illustrate its efficacy in enhancing model performance over manual tuning or random search. Grid Search searches all combinations of pre-specified hyperparameter values systematically and offers a sound estimate of their effect on model performance metrics such as AUC or accuracy. This systematic procedure is supported by statistical warranties in terms of locating the best configuration in the explored space. Even though it is computationally costly on large datasets, Grid Search's exhaustive search yields great insight into interactions between hyperparameters and their effect on model behavior, aiding a scientific understanding of model optimization procedures^{70,71}.

Algorithmic steps of optimized XGBoost

Optimized sequential neural network (OSNN)

Sequential CNNs, not solely confined to the image realm, have emerged as powerhouses for processing diverse sequential data like text, music, and time series⁷². They differ from traditional neural networks in their ability to remember and utilize information from previous elements in the sequence, allowing them to model complex temporal relationships.

In the world of cybersecurity, where nefarious forces hide in the shadows, conventional intrusion detection tends to be akin to static sentinels guarding against familiar dangers^{73–75}. Yet in the ever-changing environment of contemporary cyberattacks, such stodgy defenses are readily evaded. We tuned the sequential neural network with changing variations in convolution block, flatten and dense layer and regularization block by modifying the filter size, kernel size, dropout and various activation functions to extract the multiparametric, complex and concealed dynamics from various cyber-attacks. This is where OSNNs come into play, being nimble detectives that can decode the complex patterns concealed in the dynamic network traffic flow. Consider network traffic as a rich story, filled with subtle hints scattered along the course of events. OSNNs, in contrast to static approaches, have the extraordinary capacity to understand the temporal relations of this story^{76–79}. They can examine the timing and sequence of events, piecing together the narrative like an old detective. This renders them well-suited to detect latent patterns that may shout “intrusion!” to their learned algorithms. In addition, OSNNs have a long-term memory, capable of recalling vital information from far-flung sections of the network traffic stream^{80–82}. This is similar to a detective carefully examining past case records, leaving no stone unturned. This long memory is especially useful in detecting advanced attacks that develop over a period of time, braiding their malicious purposes through several incidents. The capabilities of SNNs go beyond pattern detection. They can learn detailed patterns from network data, similarly to how a detective learns the fine tells of an experienced con man. They can thus detect subtle anomalies that would go unnoticed by conventional rule-based systems, usually oblivious to these fine-grained variations from normality^{83,84}. The calculation of convolution layers, pooling layers, objective & activation functions, loss function and regularization methods was carried out as outlined in^{85–87}.

Pseudocode for OSNN

Resources requirements

All experiments were conducted on a MacBook Pro 2020 equipped with 8GB of RAM and accelerated by two GPUs (GPU4 × 2) to expedite the training process. This hardware configuration facilitated the efficient training of optimized Sequential Neural Networks (OSNNs) with diverse architectures, including variations in filter sizes, layer depths, kernel sizes, and batch sizes.

Results

In the ever-expanding labyrinth of the IoT where devices whisper data and secrets in a constant hum, lurks the invisible menace of cyber-attacks. This research unveils a revolutionary AI-powered framework that pulverizes existing intrusion detection barriers, safeguarding the interconnected future. We meticulously refine the data preparation battlefield, ensuring every anomaly stands out like a beacon against the noise. This AI warrior, meticulously honed, slashes through subtle attack patterns with unparalleled precision. A multi-layered fortress equipped with customizable defenses, it captures diverse attack signatures before they breach the gates. We employ the twin shields of regularization and cross-validation, safeguarding against data imbalances and ensuring real-world effectiveness.

The Fig. 4 reflect the confusion matrix yielded using logistic regression and optimized XGBoost with different settings.

Table 3 presents the results of binary classification for intrusion detection using optimized XGBoost models with various parameter settings. All XGBoost models (1, 2, and 3) significantly outperform Logistic Regression across all metrics, demonstrating their superior ability to accurately detect intrusions. Hyperparameter tuning contributes to further improvements. XGBoost Model 3, with the most extensive hyperparameter tuning, achieves higher scores compared to Models 1 and 2, suggesting that tuning can lead to incremental gains. The

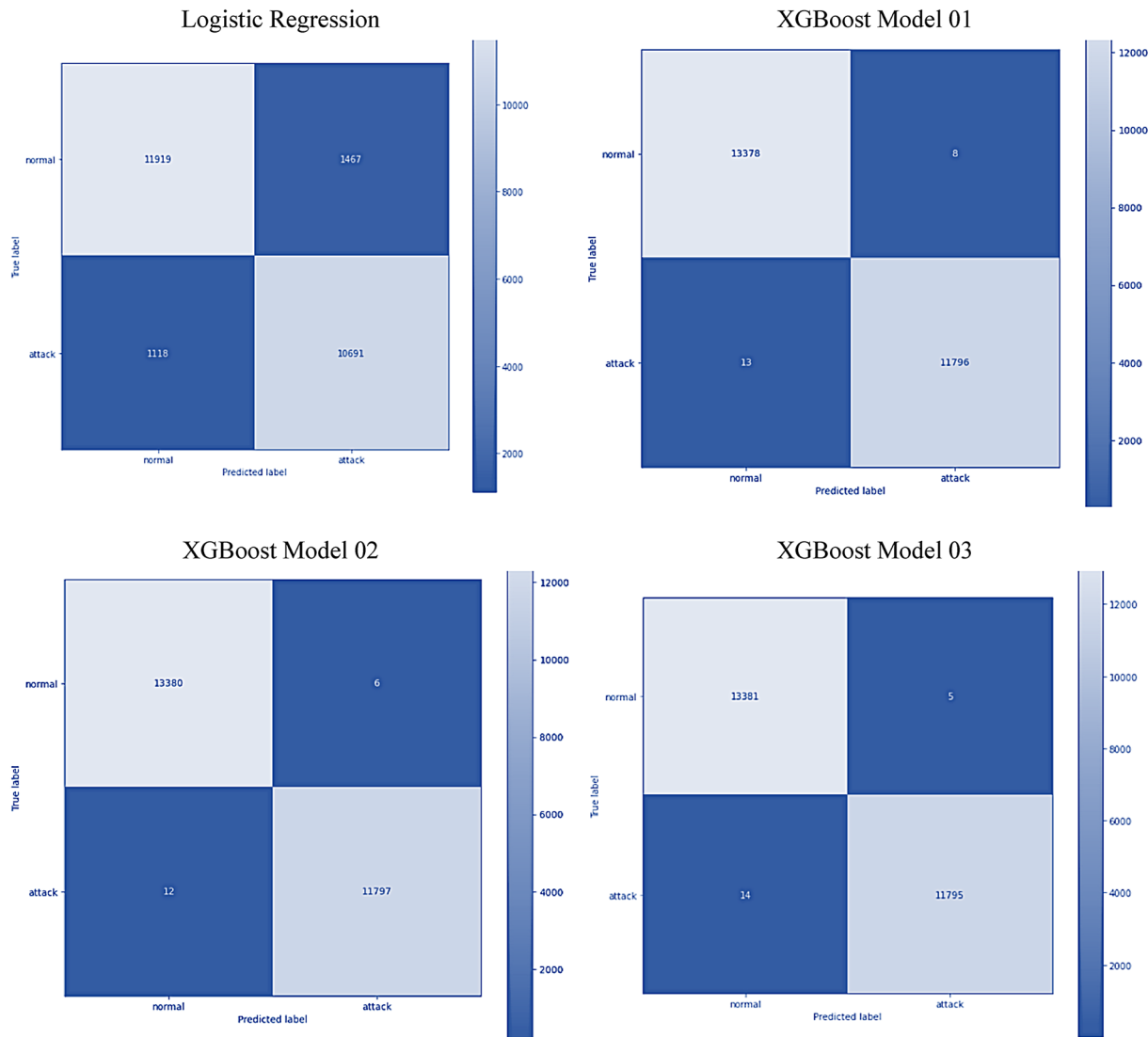


Fig. 4. Confusion matrix of Binary classification to detect intrusion attacks using Optimized XGBoost or NSL-KDD dataset.

Model	Sensitivity/TPR	Specificity	FPR	PPV	NPV	Accuracy	F1-score	MCC
Logistic Reg.	0.9142	0.8793	0.1207	0.8904	0.9053	0.8974	0.8248	0.7947
XGB-model 1	0.9990	0.9993	0.0007	0.9994	0.9989	0.9992	0.9982	0.9983
XGB-model 2	0.9990	0.9996	0.0005	0.9996	0.9988	0.9992	0.9982	0.9985
XGB-model 3	0.9991	0.9995	0.0004	0.9996	0.9990	0.9993	0.9984	0.9986

Table 3. Intrusion detection using different settings of hyperparameters of XGBoost or NSL-KDD dataset.

XGBoost models consistently exhibit high scores for all metrics, demonstrating their effectiveness in correctly identifying intrusions (high sensitivity), accurately classifying normal traffic (high specificity), minimizing false positives (high PPV), correctly identifying normal instances (high NPV), balancing sensitivity and precision (high F1-score), and demonstrating overall performance and robustness to class imbalance (high MCC).

The proposed model’s capabilities enable it to capture intricate patterns in intrusion data, leading to superior performance. Moreover, XGBoost’s regularization techniques help prevent overfitting, promoting generalization to unseen data. The XGBoost’s computational efficiency makes it suitable for large-scale intrusion detection tasks.

As shown in Table 4, SNN model 01, using its default parameters, obtained an overall accuracy of 0.99 on the test data, demonstrating exceptional performance in classifying multiclass intrusion types. The model exhibited excellent performance in detecting Normal, DoS, and U2R classes, achieving perfect detection (all performance

	Precision	Recall	F1-Score	Support
Normal	0.99	1.00	1.00	9181
DOS	0.97	0.96	0.97	2357
Probe	0.88	0.73	0.80	224
R2L	0.00	0.00	0.00	11
U2R	0.99	0.99	0.99	13,422
Accuracy			0.99	25,195
Macro Avg.	0.77	0.74	0.75	25,195
Weighted Avg.	0.99	0.99	0.99	25,195

Table 4. Multiclass intrusion detection on test data using SNN model 01 (Total parameters 14737, trainable parameters 14737) or NSL-KDD dataset.

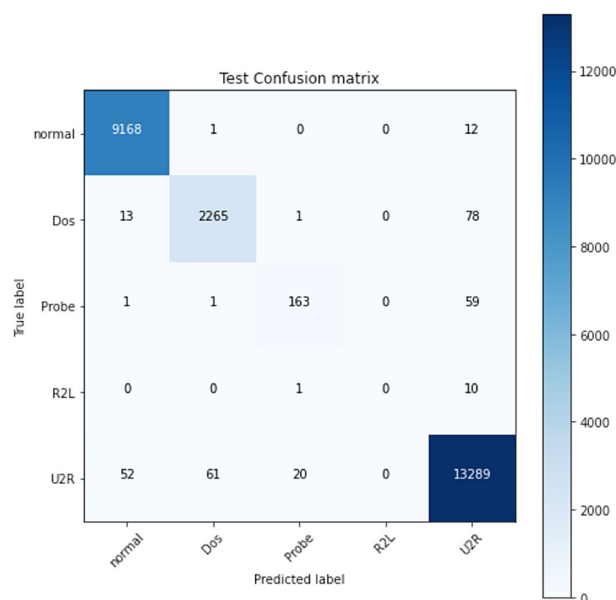


Fig. 5. Confusion matrix for multiclass (normal, DoS, Probe, R2L, L2R) intrusion detection using SNN with 14737 parameters or NSL-KDD dataset.

metrics at 1.00) for Normal subjects. Normal, DOS and U2R yielded performance greater than 97% indicating higher intrusion detection.

Figure 5 illustrates the confusion matrix for the sequential neural network, where R2L exhibits the lowest detection rate.

Figure 6 displays the loss curve for both testing and training data after 100 epochs, using the same settings as the SNN model.

The Fig. 7 presents the confusion matrix of multiclass using optimized SNN model on test data.

The Table 5 reflect the optimal hyperparameters of OSNN model.

As shown in Table 6, SNN model 02, using its optimized parameters, obtained an overall accuracy of 0.99 on the test data, demonstrating exceptional performance in classifying multiclass intrusion types. The model exhibited excellent performance in detecting Normal, DoS, and U2R classes, achieving perfect detection (all performance metrics at 1.00) for Normal subjects. For DoS attacks, Precision, Recall, and F1-Score were close to 0.97, indicating strong performance. The model also demonstrated perfect detection for U2R attacks with all scores at 1.00. However, the model exhibited weaker performance in detecting R2L intrusions, with Precision at 0.50, Recall at 0.09, and F1-Score at 0.15. This low performance is likely attributed to the limited number of R2L examples (only 11) within the dataset, which significantly impacted the macro-average scores.

Figure 8 illustrates the loss curve for training and testing data when classifying multiclass using the OSNN model. The training data achieved a loss of 0.0183, while the testing data yielded a loss of 0.01826.

As depicted in Fig. 9, the AUC scores on the test data for multiclass classification were as follows: Normal (1.00), DoS (0.98), Probe (0.86), R2L (0.50), and U2R (0.99). The model demonstrated exceptional discrimination for Normal, DoS, Probe, and U2R classes, suggesting a strong ability to distinguish these attacks from normal traffic.

SNN Model 01, with 14,737 trainable parameters, and SNN Model 02, with 152,580 trainable parameters, exhibit varying levels of architectural complexity. This complexity, achieved through adjustments in filter sizes,

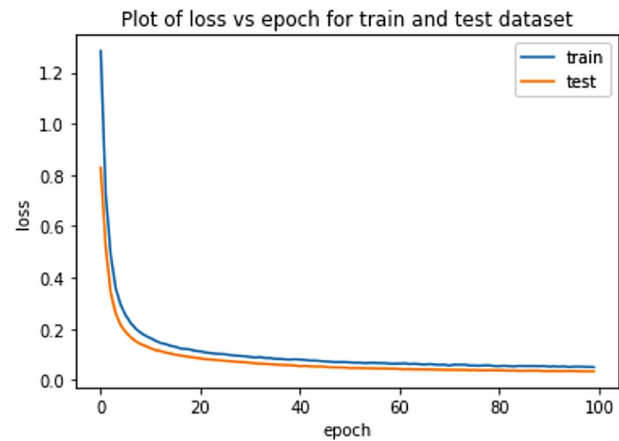


Fig. 6. Loss curve for multiclass (normal, DoS, Probe, R2L, L2R) intrusion detection using SNN with 14,737 parameters.

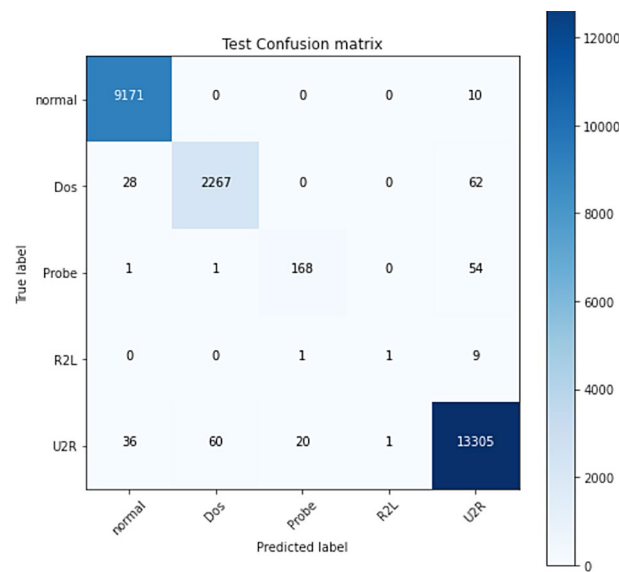


Fig. 7. Confusion matrix for multiclass (normal, DoS, Probe, R2L, L2R) intrusion detection using OSNN with 152,580 parameters or NSL-KDD dataset.

Parameters List	Optimal values
Solver_name	SGDM
Kernel size	7
L2-regularization	0.0001
Dense unit	150
Fully_connected_layers	2
Initial_learning_rate	Single: 0.0001
Loss_function	Cross Entropy
Dropout_rate (%)	0.4

Table 5. Optimal parameters of optimized SNN model.

kernel sizes, and the number of dense layers, enables the models to effectively capture intricate patterns within the highly complex intrusion datasets. Model predictions are influenced by the interplay of all parameters, including the weights and biases of neurons. These parameters, dynamically learned during training, shape the model's decision boundaries. Furthermore, careful optimization of hyperparameters, including the selection of

	Precision	Recall	F1-Score	Support
Normal	0.99	1.00	1.00	9181
DOS	0.97	0.96	0.97	2357
Probe	0.89	0.75	0.81	224
R2L	0.50	0.09	0.15	11
U2R	1.00	1.00	1.00	13,422
Accuracy			0.99	25,195
Macro Avg.	0.87	0.76	0.78	25,195
Weighted Avg.	0.99	0.99	0.99	25,195

Table 6. Multiclass intrusion detection on test data using SNN model 2 with optimized parameters or NSL-KDD dataset with 152,580 parameters.

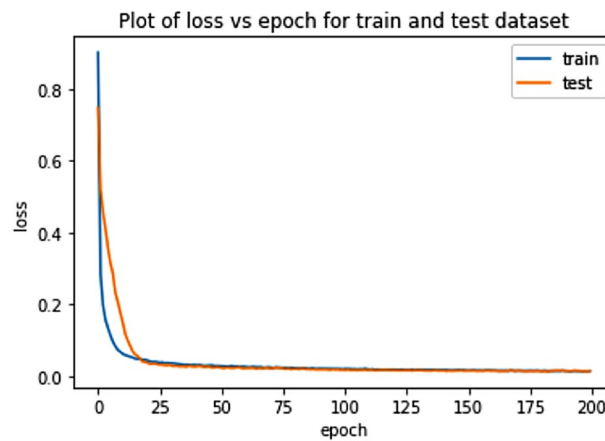


Fig. 8. Loss curve for multiclass (normal, DoS, Probe, R2L, L2R) intrusion detection using OSNN with 152,580 parameters or NSL-KDD dataset.

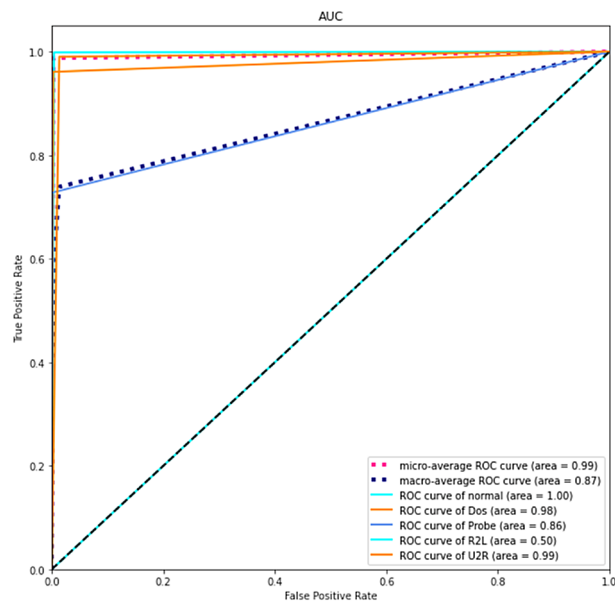


Fig. 9. AUC curve for multiclass (normal, DoS, Probe, R2L, L2R) intrusion detection using OSNN with 152,580 parameters for NSL-KDD dataset.

Authors	Methods	Findings
88	MCA-LSTM	Accuracy: 88.11% False Alarm rate (FAR): 29.62%
89	RNN	83.28%
90	FFNN-LSO	Accuracy: 95.24% FAR: 9.40%
	FFNN-GA	Accuracy: 86.44% FAR: 29.1%
	FFNN-PSO	Accuracy: 94.42% FAR: 14.78%
This study	Optimized SNN	Accuracy: 96.80% FAR: 7.30%

Table 7. Comparision of intrusion detetion performance on UNSW-NB15 dataset with existing studies.

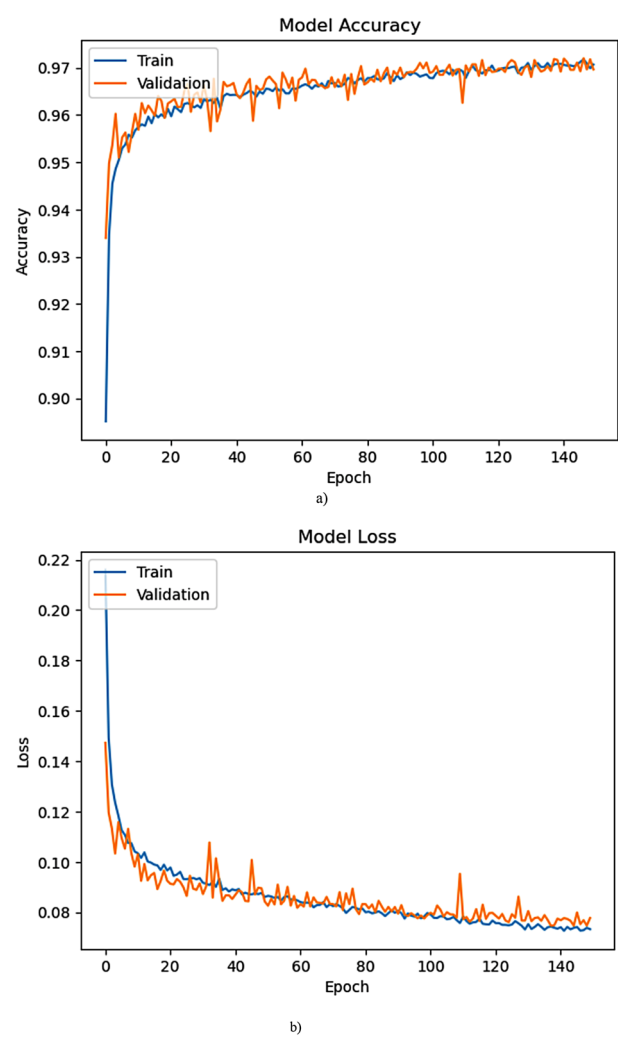


Fig. 10. Accuracy Loss curve for intrusion detection using Optimized SNN on UNSW-NB15 dataset.

activation functions (ReLU, GeLU, LeakyReLU), tuning of the learning rate, and implementation of dropout and regularization techniques, significantly enhances the models' ability to accurately classify different types of intrusions. The Table 7 achived improved detection accuracy of 96.80% for intrusion detection on UNSW-NB15 dataset.

Figure 10 depicts the loss curves for training and testing data during multi-class classification using the Optimized SNN model on the UNSW-NB15 dataset. The training data achieved a final loss of 0.0733, while the testing data exhibited a slightly higher loss of 0.0777. This small discrepancy between training and testing loss suggests that the model may be generalizing well, with minimal overfitting observed.

Authors	Methods	Findings
91	LSTM-AE	Accuracy: 99.10% Loss: 0.0040 Recall: 99.10% Precision: 99.07% F-Measure: 99.02%
7	CNN-GRU	Accuracy: 98.73%
92	Generative Adversarial Network (GAN)	Accuracy: 98.48% Precision: 97.27% Recall: 94.25 F1-Score: 95.56%
This study	Optimized SNN	Accuracy: 99.53% Sensitivity: 99.69% Specificity: 99.24% PPV: 99.56% NPV: 99.46% F1-Score: 99.31%

Table 8. Comparison of intrusion detetion performance for CSE-CICIDS-2018 dataset with existing studies.

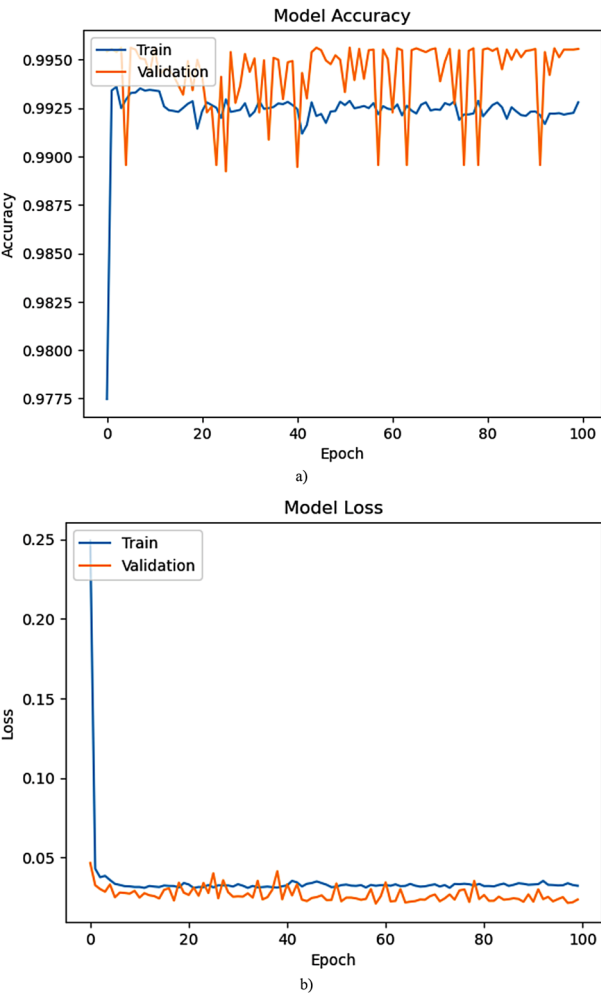


Fig. 11. Accuracy Loss curve for intrusion detection using Optimied SNN on CSE-CICIDS18 dataset.

The Table 8 reflects the performance of detecting intrusion based on CSE-CICIDS-2018 dataset. The proposed model yielded the optimal performance with test accuracy of 99.53% using OSNN.

Figure 11 presents the loss curves for the training and testing phases of an Optimized SNN model performing multi-class classification on the CSE-CICIDS18 dataset. The training loss converges to a value of 0.0323, indicating effective learning on the training data. The testing loss, while slightly higher at 0.0236, remains close to the training loss. This small discrepancy suggests that the model exhibits good generalization capabilities with minimal overfitting.

Discussions

This research rigorously evaluated the proposed intrusion detection system on three challenging datasets: NSL-KDD, CICIDS2017 and UNSW-NB15. The results depicts effectiveness of optimized XGBoost and OSNN models in detecting various types of intrusions.

Table 9 showcases a diverse range of datasets (NSL-KDD, CIDDS-001, UNSW-NB15, KDD Cup'99, CICIDS2017) and machine learning models (ML, DL, DNN, ANN, RNN, LSTM, GRU, XGBoost, SVM, KNN, RF, GBM, PCA, PSO, FFDNNs, GA-LR, Ensemble methods) employed in intrusion detection research. The research reveals effectiveness of various machine learning techniques, with many studies reporting high accuracy rates exceeding 90% across different datasets.

The study in row⁹⁹ highlights the effectiveness of ensemble methods in improving detection accuracy, emphasizing the benefits of combining multiple models to enhance performance. The authors in⁹⁵ employed feature selection methods like FFDNNs to improve accuracy while acknowledging their potential computational complexity. The work in² specifically states that computational complexity is a key consideration, with emphasis on the need for a balance between accuracy and efficiency, particularly with big data. Authors in⁹⁴ emphasize the prospect of hybrid models that merge legacy machine learning (XGBoost) and deep learning (LSTM) as evidence of increasing interest in combining the strengths of both approaches.

These findings collectively indicate the effectiveness of several machine learning techniques, highlight the growing relevance of deep learning, highlight the importance of feature selection and ensemble methods, and validate the importance of considering computational complexity in intrusion detection systems.

In studies¹⁰⁰, the efficacy of different deep learning models, i.e., LSTM, Simple RNN, and CNN-LSTM, in intrusion detection using the NSL-KDD dataset was evaluated. Preprocessing involved scaling input to deal with variability in attribute values. The CNN-LSTM model¹⁰¹ demonstrated the highest accuracy, yielding 79.37% for binary classification and 70.13% for multi-classification, surpassing the performance of LSTM (74.77% and 68.78%) and Simple RNN (68.55% and 64.67%). The authors concluded that the CNN-LSTM architecture showed promise for intrusion detection and that further improvements could be achieved by incorporating more sophisticated feature extraction methods. Z. Li and Z.A. Qin⁹⁶ proposed an IDS that leverages a semantic-based algorithm to parse input data, followed by LSTM for prediction. Utilizing the NSL-KDD dataset and evaluating on test accuracy and F1-score, their Semantic-LSTM model achieved 82.21% accuracy and 82.27% F1-score, outperforming traditional methods. While promising, the authors acknowledged the need for further exploration of other RNN variants. The authors⁹⁷ investigated the effectiveness of LSTM and GRU recurrent neural networks for intrusion detection, evaluating their performance on the NSL-KDD, BCNET, and BGW datasets. With the metrics of test accuracy and F1-score, the research discovered that on the NSL-KDD dataset, the LSTM model had 82.78% test accuracy and 83.34% F1-score, and the GRU model had 82.87% accuracy and 83.05% F1-score. The researchers⁹⁸ tested many deep learning hybrid methods for intrusion detection using the NSL-KDD dataset. The LSTM architecture, among others, was tested using test accuracy, FPR, and FNR as measuring metrics for performance. The best-performing LSTM configuration yielded an FPR of 3.00%, an FNR of 26.00%, and an accuracy of 82.00%.

The research presented an IDS with a Genetic Algorithm (GA) for feature selection and Logistic Regression (LR) as the classifier Evaluated on KDDCup99 and UNSW-NB15, the GA-LR identified 18 and 20 optimal features respectively. Using accuracy and False Alarm Rate as metrics, the GA-LR achieved 99.90% accuracy and 0.105% FAR on KDDCup99, and 81.42% accuracy and 6.39% FAR on UNSW-NB15. In contrast, study²⁴ employed a Gain Ratio (GR) based feature selection method with an Artificial Neural Network (ANN-MLP) classifier on the UNSW-NB15 dataset. The GR method selected 30 features, resulting in an ANN-MLP model with 76.96% accuracy and 79.80% precision.

The Table 10 presents the comparative results with existing studies. The proposed approach, combining optimized XGBoost for binary classification and optimized sequential neural networks for multiclass

Author	Dataset	Methods	Results
2	NSL-KDD, UNSW-NB15, CIDDS-001,	Machine Learning, PCA, RNN, PSO, SVM, KNN	Validation accuracy of: UNSW-NB15 (98.53%), NSL-KDD (94.58%), CIDDS-001 (97.87%)
3	CIDDS-001, NSLKDD	CNN, RF, DNN, DL, RNN, ML, ML classifier stacking	Accuracy of 97.99%
93	UNSWNB15, CICIDS2017, KDD Cup'99 and NSL-KDD	Xgboost, RF, K-NN, ANN, SVM, ML RNN, DNN,, DL	Above 80% accuracy using DNN and ANN algorithms.
94	NSL-KDD and UNSW-NB15	XGBoost, RNN, LSTM, ML, and GRU	85.14% accuracy
95	KDDCup'99dataset for NSL-KDD	ML, Decision Tree, NB, SVM KNN, DL, FFDNNs	86.53% accuracy
96	NSL-KDD	LSTM	82.21% accuracy
97	NSL-KDD, BCNET, and BGW	LSTM and GRU	82.87% accuracy
98	NSL-KDD	LSTM, DNN	75.75% accuracy

Table 9. Comparison of results with existing studies of different intrusion detection dataset for binary and multiclass classification.

Author	Method	Binary Accuracy	Multiclass accuracy
Tang et al ¹⁰² .	DNN	77.75%	
Hosseini ¹⁰³ ,	PSO-ANN	88.90%	
Hosseini ¹⁰³ ,	GA-ANN	83.11%	
Su et al ¹⁰⁴ .	BAT	82.56%	
Su et al ¹⁰⁴ .	BAT-MC	84.20%	
Gao et al. [105] ¹⁰⁵	MultiTree	85.20%	
Zhang et al ¹⁶ .	AE-DNN	--	79.74%
Hu et al ¹⁰⁷ .	AS-CNN	--	80.00%
Tama et al ¹⁰⁷ .	TSE-IDS	85.79%	--
Bostani, and Sheikhan ¹⁰⁸	BGSA-MI	88.36%	--
Bostani, and Sheikhan ¹⁰⁸	Relief	84.60%	--
Almasoudy et al ¹⁰⁹ .	DE-ELM	87.83%	80.15%
Sarumi et al ¹¹⁰ .	Filter SVM	--	77.17%
This Study	Optimized XGBoost Optimized Sequential Neural network	99.93%	99.00%

Table 10. Comparison of intrusion detection with existing studies for NSL-KDD dataset.

classification, significantly outperforms all other methods listed in both binary accuracy (99.93%) and multiclass accuracy (99.00%). This substantial improvement demonstrates its remarkable potential for intrusion detection. The accuracy gains over existing methods are substantial, with improvements of over 10% in both binary and multiclass accuracy. This suggests the potential for more robust and reliable intrusion detection systems. Table 4 highlights the diversity of approaches applied to intrusion detection, including deep DNN, particle swarm optimization-based ANN PSO-ANN, GA-ANN, BAT, and more. This underscores the ongoing exploration and innovation in this field. The optimization of both XGBoost and the neural network architecture likely contributes significantly to the improved performance. The high accuracy scores across both binary and multiclass tasks suggest that the proposed approach might generalize well to various intrusion detection scenarios.

The aforementioned datasets effectively capture the complex and often subtle dynamics inherent in intrusion detection, necessitating the development of advanced tools to enhance predictive capabilities. This study investigates the potential of ML and DL algorithms, with a particular focus on vision transformers, for improving intrusion detection. Our findings demonstrate the robustness of this approach in detecting anomalies within network traffic.

In an effort to combat the computational complexity and excessive false detection rates typical of existing intrusion detection models, this paper recommends a sequence of groundbreaking improvements. These are exhaustive data preprocessing—normalization, balancing, encoding, and feature extraction—to achieve high-quality input for the model. A fine-tuned XGBoost model, additionally optimized using deep grid search, attains optimal detection accuracy. Moreover, a stronger Sequential Neural Network (SNN) is built with optimally selected hyperparameters, activation functions, and dropout layers to improve flexibility and generalizability. Finally, L2 regularization and 5-fold cross-validation are used to prevent overfitting and enhance model robustness, thereby providing an effective countermeasure against adaptive IoT cyberattacks. The model’s credibility is ensured by its high accuracy on diverse sets of datasets (NSL-KDD, UNSW-NB15, and CICIDS2017), minimal false positives, and high generalization capability. The credibility of the OSNN model is also ensured by proper hyperparameter fine-tuning, for example, with the use of stable activation functions and optimal parameters for learning. Its effectiveness in real-world tests and adaptive architecture ensures its applicability even in evolving threats.

Conclusion

In the rapidly evolving and data-driven environment of the Internet of Things (IoT), robust security measures are required to counter the growing complexity of cyber attacks. Machine learning-based anomaly detection offers a promising solution by identifying unusual network behavior indicative of malicious intrusion. However, existing intrusion detection systems are prone to fail in addressing the dynamic and complex nature of modern threats.

In this research, it is suggested a new Intrusion Detection System (IDS) taking advantage of improved XGBoost and Sequential Neural Network (SNN) models to overcome shortcomings of current techniques. Earlier works usually encountered hardships in computational intricacy, ill-prepared preprocessing of data, and limited options for optimization techniques. To address these issues, this paper incorporates several key enhancements, i.e., meticulous preprocessing of data approaches, comprehensive set of measurement parameters, and optimized XGBoost and OSNN model designs. The system was tested rigorously using three real-world datasets: NSL-KDD, UNSW-NB15, and CICIDS2017. On NSL-KDD dataset, the XGBoost model was top-notch with an accuracy of 99.93%, F1-score of 99.84%, Matthews Correlation Coefficient (MCC) of 99.86%, and very low False Positive Rate (FPR) of 0.0004. Optimized SNN model was also high-performing on NSL-KDD with an accuracy of 99.0% and an Area Under the Curve (AUC) of 1.00. Moreover, the OSNN model was better on the UNSW-NB15 dataset, where it achieved an accuracy of 96.80% and a loss of 0.0777. On the CICIDS-2017 dataset, the OSNN model achieved a high accuracy of 99.53% and a loss of 0.0236. The OSNN model is supported by meticulous

hyperparameter tuning, including the use of stable activation functions (ReLU, GeLU, LeakyReLU), learning rate tuning, and the use of dropout and regularization techniques. These adjustments enable the model to learn successful complex intrusion patterns, resulting in dramatic accuracy improvement on diverse and challenging datasets.

In general, the demonstrated excellence of our new model, by its high accuracy, low false positive rates, and outstanding generalization to different datasets, unequivocally demonstrates its potential as a powerful tool to enhance intrusion detection performance. This success verifies the efficacy of synergistically combining optimized DL and ML techniques to address the complex and dynamic issues in protecting the IoT space. For the future, these results offer a rich array of promising avenues for further investigation. The flexible nature of the model enables constant adjustment and extension, enabling the addition of new attack vectors and shifting IoT device dynamics. Future work can explore real-time and edge-deployment-based horizons, facilitating the preventive interception of threats at their origin. Moreover, the capability of the model to maintain system integrity, prevent fraud, and optimize network performance carries over from intrusion detection into broader uses in other areas of IoT security, including anomaly detection in critical infrastructure and tailored security for smart homes. Finally, this research not only improves the security position of current IoT networks but also provides a good foundation for the development of smart and adaptive security systems that can respond very well to the constantly evolving nature of the threat landscape.

Limitations and future directions

The present assessment, which is mainly conducted using proven datasets, might not entirely reflect the variety and intricacy of actual IoT settings, which is a limitation to this study. In addition, the computational expense of deep learning models, while minimized, can still pose an issue for resource-limited IoT devices. Moreover, the interpretability of intricate deep learning models can be restricted, making it difficult to instill trust and confidence in their decision-making process.

Data availability

The NSL-KDD dataset is publicly available at (<https://data.mendeley.com/datasets/t5bffpjd28/1>) and (<https://www.kaggle.com/datasets/hassan06/nslkdd>) without any restrictions under license CC0 1.0 and used by [53]. UNSW-NB15 publicly available at (<https://www.kaggle.com/datasets/dhoogla/unswnb15>) under license CC BY-NC-SA 4.0 and CICIDS2017 is publicly available at (<https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset>) under licence CC0.

Received: 14 November 2024; Accepted: 9 June 2025

Published online: 01 July 2025

References

- Hussain, A. & Kumar Sharma, P. Efficient working of signature based intrusion detection technique in computer networks. *Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol.* 60–64. <https://doi.org/10.32628/CSEIT195215> (2019).
- Albahar, M. A. et al. Improving intrusion detection system using artificial neural network. *Int. J. Adv. Comput. Sci. Appl.* 11 <https://doi.org/10.14569/IJACSA.2020.0110670> (2020).
- Abdulhammed, R., Faezipour, M., Abuzneid, A. & AbuMallouh, A. Deep and machine learning approaches for Anomaly-Based intrusion detection of imbalanced network traffic. *IEEE Sens. Lett.* 3, 1–4. <https://doi.org/10.1109/LSSENS.2018.2879990> (2019).
- Sarkar, A., Sharma, H. S. & Singh, M. M. A supervised machine learning-based solution for efficient network intrusion detection using ensemble learning based on hyperparameter optimization. *Int. J. Inf. Technol.* 15, 423–434. <https://doi.org/10.1007/s41870-022-01115-4> (2023).
- Adeel, A., Larijani, H. & Ahmadiania, A. Random neural network based novel decision making framework for optimized and autonomous power control in LTE uplink system. *Phys. Commun.* 19, 106–117. <https://doi.org/10.1016/j.phycom.2015.11.004> (2016).
- Jin, X. et al. Deep learning with S-Shaped rectified linear activation units. *Proc. AAAI Conf. Artif. Intell.* 30 <https://doi.org/10.1609/aaai.v30i1.10287> (2016).
- Henry, A. et al. Composition of hybrid deep learning model and feature optimization for intrusion detection system. *Sensors* 23, 890. <https://doi.org/10.3390/s23020890> (2023).
- Wang, C. et al. Hybrid intrusion detection system based on combination of random forest and autoencoder. *Symmetry (Basel)* 15, 568. <https://doi.org/10.3390/sym15030568> (2023).
- Sajid Farooq, M. et al. A fused machine learning approach for intrusion detection system. *Comput. Mater. Contin.* 74, 2607–2623. <https://doi.org/10.32604/cmc.2023.032617> (2023).
- Mehmood, M. et al. A hybrid approach for network intrusion detection. *Comput. Mater. Contin.* 70, 91–107. <https://doi.org/10.32604/cmc.2022.019127> (2022).
- Nedeljkovic, D. & Jakovljevic, Z. CNN based method for the development of cyber-attacks detection algorithms in industrial control systems. *Comput. Secur.* 114, 102585. <https://doi.org/10.1016/j.cose.2021.102585> (2022).
- Mebawondu, J. O., Alowolodu, O. D., Mebawondu, J. O. & Adetunmbi, A. O. Network intrusion detection system using supervised learning paradigm. *Sci. Afr.* 9, e00497. <https://doi.org/10.1016/j.sciaf.2020.e00497> (2020).
- Tavallaei, M., Bagheri, E., Lu, W. & Ghorbani, A. A. A detailed analysis of the KDD CUP 99 data set. In: 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications. IEEE, pp 1–6 (2009).
- Khraisat, A., Gondal, I. & Vamplew, P. An Anomaly Intrusion Detection System Using C5 Decision Tree Classifier. pp 149–155 (2018).
- Ever, Y. K., Sekeroglu, B. & Dimililer, K. Classification Analysis of Intrusion Detection on NSL-KDD Using Machine Learning Algorithms. pp 111–122 (2019).
- Zhang, C. et al. A Deep Learning Approach for Network Intrusion Detection Based on NSL-KDD Dataset. In: 2019 IEEE 13th International Conference on Anti-counterfeiting, Security, and (ASID). IEEE, pp 41–45 (2019).
- Sperotto, A. et al. An overview of IP Flow-Based intrusion detection. *IEEE Commun. Surv. Tutorials* 12, 343–356. <https://doi.org/10.1109/SURV.2010.032210.00054> (2010).
- Smamarwar, S. K., Gupta, G. P., Kumar, S. & Kumar, P. An optimized and efficient android malware detection framework for future sustainable computing. *Sustain. Energy Technol. Assessments* 54, 102852. <https://doi.org/10.1016/j.seta.2022.102852> (2022).

19. Smmarwar, S. K., Gupta, G. P. & Kumar, S. Android malware detection and identification frameworks by leveraging the machine and deep learning techniques: A comprehensive review. *Telemat Inf. Rep.* **14**, 100130. <https://doi.org/10.1016/j.teler.2024.100130> (2024).
20. Smmarwar, S. K., Gupta, G. P. & Kumar, S. A hybrid feature selection Approach-Based android malware detection framework using machine learning techniques. In *Cyber Security, Privacy and Networking: Proceedings of ICSPN 2021* 347–356 (Singapore: Springer Nature Singapore, 2022).
21. Smmarwar, S. K., Gupta, G. P. & Kumar, S. Design of a fused triple convolutional neural network for malware detection: A visual classification approach. In *Advances in Computing and Data Sciences: 5th International Conference, ICACDS 2021*, Revised Selected Papers, Part I 5 2021, 279–289 (Springer International Publishing, Nashik, India, 2021).
22. Smmarwar, S. K., Gupta, G. P. & Kumar, S. Malware Detection Framework Based on Iterative Neighborhood Component Analysis for Internet of Medical Things. pp 98–106 (2024).
23. Smmarwar, S. K., Gupta, G. P., Kumar, S., Intelligence & Computing XAI-AMD-DL: An Explainable AI Approach for Android Malware Detection System Using Deep Learning. In: 2023 IEEE World Conference on Applied and (AIC). IEEE, pp 423–428 (2023).
24. Smmarwar, S. K., Gupta, G. P. & Kumar, S. Deep malware detection framework for IoT-based smart agriculture. *Comput. Electr. Eng.* **104**, 108410. <https://doi.org/10.1016/j.compeleceng.2022.108410> (2022).
25. Smmarwar, S. K., Gupta, G. P. & Kumar, S. AI-empowered malware detection system for industrial internet of things. *Comput. Electr. Eng.* **108**, 108731. <https://doi.org/10.1016/j.compeleceng.2023.108731> (2023).
26. Alqahtani, H. & Kumar, G. Deep learning-based intrusion detection system for in-vehicle networks with knowledge graph and statistical methods. *Int. J. Mach. Learn. Cybern.* <https://doi.org/10.1007/s13042-024-02465-0> (2024).
27. Geeta Kocher, G. & Kumar A HYBRID DEEP LEARNING APPROACH FOR EFFECTIVE INTRUSION DETECTION SYSTEMS USING SPATIAL-TEMPORAL FEATURES. *Adv. Eng. Sci.* **54**, 1503–1519 (2022).
28. Lo, W. et al. A hybrid deep learning based intrusion detection system using spatial-temporal representation of in-vehicle network traffic. *Veh. Commun.* **35**, 100471. <https://doi.org/10.1016/j.vehcom.2022.100471> (2022).
29. Kumar, G. & Alqahtani, H. Machine learning techniques for intrusion detection systems in SDN-Recent advances, challenges and future directions. *Comput. Model. Eng. Sci.* **134**, 89–119. <https://doi.org/10.32604/cmes.2022.020724> (2023).
30. Wu, Z., Zhang, H., Wang, P. & Sun, Z. RTIDS: A robust Transformer-Based approach for intrusion detection system. *IEEE Access.* **10**, 64375–64387. <https://doi.org/10.1109/ACCESS.2022.3182333> (2022).
31. Dr, S., Smys, D. A., Basar, D. H. & Wang Hybrid intrusion detection system for internet of things (IoT). *J. ISMAC.* **2**, 190–199. <https://doi.org/10.36548/jismac.2020.4.002> (2020).
32. Abdel-Basset, M., Hawash, H., Chakraborty, R. K. & Ryan, M. J. Semi-Supervised Spatiotemporal deep learning for intrusions detection in IoT networks. *IEEE Internet Things J.* **8**, 12251–12265. <https://doi.org/10.1109/JIOT.2021.3060878> (2021).
33. Yang, L. & Shami, A. A Transfer Learning and Optimized CNN Based Intrusion Detection System for Internet of Vehicles. In: ICC 2022 - IEEE International Conference on Communications. IEEE, pp 2774–2779 (2022).
34. Toldinas, J. et al. A novel approach for network intrusion detection using multistage deep learning image recognition. *Electronics* **10**, 1854. <https://doi.org/10.3390/electronics10151854> (2021).
35. Xu, Z. & Liu, M. An Intrusion Detection Method based on PCA and ResNet. In: 2021 3rd International Conference on Applied Machine Learning (ICAML). IEEE, pp 52–56 (2021).
36. Xiao, Y. & Xiao, X. An intrusion detection system based on a simplified residual network. *Information* **10**, 356. <https://doi.org/10.3390/info10110356> (2019).
37. Hoang, T.-N. & Kim, D. Supervised contrastive ResNet and transfer learning for the in-vehicle intrusion detection system. *Expert Syst. Appl.* **238**, 122181. <https://doi.org/10.1016/j.eswa.2023.122181> (2024).
38. Shaikh, A. & Gupta, P. Real-time intrusion detection based on residual learning through ResNet algorithm. *Int. J. Syst. Assur. Eng. Manag.* <https://doi.org/10.1007/s13198-021-01558-1> (2022).
39. Kodys, M., Lu, Z., Fok, K. W. & Thing, V. L. L. Intrusion Detection in Internet of Things using Convolutional Neural Networks. In: 2021 18th International Conference on Privacy, Security and Trust (PST). IEEE, pp 1–10 (2021).
40. Hussain, F. et al. IoT DoS and DDoS Attack Detection using ResNet. In: 2020 IEEE 23rd International Multitopic Conference (INMIC). IEEE, pp 1–6 (2020).
41. Song, J. et al. ReAL: A New ResNet-ALSTM Based Intrusion Detection System for the Internet of Energy. In: 2020 IEEE 45th Conference on Local Computer Networks (LCN). IEEE, pp 491–496 (2020).
42. Khoei, T. T., Hu, W. C. & Kaabouch, N. Residual Convolutional Network for Detecting Attacks on Intrusion Detection Systems in Smart Grid. In: 2022 IEEE International Conference on Electro Information Technology (eIT). IEEE, pp 231–237 (2022).
43. Om Kumar, C. U., Marappan, S., Murugesan, B. & Beaulah, P. M. R. Intrusion detection model for IoT using recurrent kernel convolutional neural network. *Wirel. Pers. Commun.* **129**, 783–812. <https://doi.org/10.1007/s11277-022-10155-9> (2023).
44. Alkadi, O., Moustafa, N., Turnbull, B. & Choo, K.-K. R. A deep blockchain Framework-Enabled collaborative intrusion detection for protecting IoT and cloud networks. *IEEE Internet Things J.* **8**, 9463–9472. <https://doi.org/10.1109/JIOT.2020.2996590> (2021).
45. Kaya, Ç., Yıldız, O. & Ay, S. Performance analysis of machine learning techniques in intrusion detection. In: 2016 24th Signal Processing and Communication Application Conference (SIU). IEEE, pp 1473–1476 (2016).
46. Shi, W.-C. & Sun, H.-M. DeepBot: a time-based botnet detection with deep learning. *Soft Comput.* **24**, 16605–16616. <https://doi.org/10.1007/s00500-020-04963-z> (2020).
47. AL-Nomasy, N., Alamri, A., Aljuhani, A. & Kumar, P. Transformer-based knowledge distillation for explainable intrusion detection system. *Comput. Secur.* **154**, 104417. <https://doi.org/10.1016/j.cose.2025.104417> (2025).
48. Kumar, P., Javeed, D., Islam, A. K. M. N. & Luo, X. (eds) (Robert) DeepSecure: A computational design science approach for interpretable threat hunting in cybersecurity decision making. *Decis Support Syst* 188:114351. (2025). <https://doi.org/10.1016/j.dss.2024.114351>
49. Kumar, P., Jolfaei, A. & Najmul Islam, A. K. An enhanced Deep-Learning empowered Threat-Hunting framework for software-defined internet of things. *Comput. Secur.* **148**, 104109. <https://doi.org/10.1016/j.cose.2024.104109> (2025).
50. Dey, A. K., Gupta, G. P. & Sahu, S. P. An efficient cyber assault detection system using feature optimization for IoT-based cyberspace. *Procedia Comput. Sci.* **235**, 757–766. <https://doi.org/10.1016/j.procs.2024.04.072> (2024).
51. Catillo, M., Pecchia, A. & Villano, U. MultiCIDS: Anomaly-based collective intrusion detection by deep learning on iot/cps multivariate time series. *Internet Things.* **30**, 101519. <https://doi.org/10.1016/j.iot.2025.101519> (2025).
52. Fares, I. A. et al. TFKAN: transformer based on Kolmogorov–Arnold networks for intrusion detection in IoT environment. *Egypt. Inf. J.* **30**, 100666. <https://doi.org/10.1016/j.eij.2025.100666> (2025).
53. Zhou, Y., Cheng, G., Jiang, S. & Dai, M. Building an efficient intrusion detection system based on feature selection and ensemble classifier. *Comput. Networks.* **174**, 107247. <https://doi.org/10.1016/j.comnet.2020.107247> (2020).
54. Jaw, E. & Wang, X. Feature selection and Ensemble-Based intrusion detection system: an efficient and comprehensive approach. *Symmetry (Basel).* **13**, 1764. <https://doi.org/10.3390/sym13101764> (2021).
55. Sharma, S., Gigras, Y., Chhikara, R. & Dhull, A. Analysis of NSL KDD dataset using classification algorithms for intrusion detection system. *Recent. Pat. Eng.* **13**, 142–147. <https://doi.org/10.2174/1872212112666180402122150> (2019).
56. Bou-Harb, E., Debbabi, M. & Assi, C. Cyber scanning: A comprehensive survey. *IEEE Commun. Surv. Tutorials.* **16**, 1496–1519. <https://doi.org/10.1109/SURV.2013.102913.00020> (2014).

57. Upadhyay, D., Manero, J., Zaman, M. & Sampalli, S. Gradient boosting feature selection with machine learning classifiers for intrusion detection on power grids. *IEEE Trans. Netw. Serv. Manag.* **18**, 1104–1116. <https://doi.org/10.1109/TNSM.2020.3032618> (2021).
58. Ding, X. et al. Gradient boosting decision tree in the prediction of no $\times$ emission of waste incineration. *Energy* **264**, 126174. <https://doi.org/10.1016/j.energy.2022.126174> (2023).
59. Aguilar Madrid, E. & Antonio, N. Short-Term electricity load forecasting with machine learning. *Information* **12**, 50. <https://doi.org/10.3390/info12020050> (2021).
60. Ahmed, S. et al. The deep learning ResNet101 and ensemble XGBoost algorithm with hyperparameters optimization accurately predict the lung Cancer. *Appl. Artif. Intell.* **37**, 2166222. <https://doi.org/10.1080/08839514.2023.2166222> (2023).
61. Budholiya, K., Shrivastava, S. K. & Sharma, V. An optimized XGBoost based diagnostic system for effective prediction of heart disease. *J. King Saud Univ. - Comput. Inf. Sci.* **34**, 4514–4523. <https://doi.org/10.1016/j.jksuci.2020.10.013> (2022).
62. Hjort, A., Pensar, J., Scheel, I. & Sommervoll, D. E. House price prediction with gradient boosted trees under different loss functions. *J. Prop. Res.* **39**, 338–364. <https://doi.org/10.1080/09599916.2022.2070525> (2022).
63. Qiu, Y. et al. Performance evaluation of hybrid WOA-XGBoost, GWO-XGBoost and BO-XGBoost models to predict blast-induced ground vibration. *Eng. Comput.* **38**, 4145–4162. <https://doi.org/10.1007/s00366-021-01393-9> (2022).
64. Ullah, I., Rios, A., Gala, V. & McKeever, S. Explaining deep learning models for tabular data using Layer-Wise relevance propagation. *Appl. Sci.* **12**, 136. <https://doi.org/10.3390/app12010136> (2021).
65. K S NP, N. D. Improved clinical diagnosis using predictive analytics. *IEEE Access.* **10**, 75158–75175. <https://doi.org/10.1109/ACCESS.2022.3190416> (2022).
66. Priscilla, C. V. & Prabha, D. P. Influence of Optimizing XGBoost to handle Class Imbalance in Credit Card Fraud Detection. In: 2020 Third International Conference on Smart Systems and Inventive Technology (ICSSIT). IEEE, pp 1309–1315 (2020).
67. Ahmed, S. et al. The deep learning ResNet101 and ensemble XGBoost algorithm with hyperparameters optimization accurately predict the lung Cancer. *Appl. Artif. Intell.* **37** <https://doi.org/10.1080/08839514.2023.2166222> (2023).
68. Pan, S., Zheng, Z., Guo, Z. & Luo, H. An optimized XGBoost method for predicting reservoir porosity using petrophysical logs. *J. Pet. Sci. Eng.* **208**, 109520. <https://doi.org/10.1016/j.petrol.2021.109520> (2022).
69. Bloch, L. & Friedrich, C. M. Using Bayesian Optimization to Effectively Tune Random Forest and XGBoost Hyperparameters for Early Alzheimer's Disease Diagnosis. pp 285–299 (2021).
70. Yu, T. & Zhu, H. *Hyper-Parameter Optimization* (A Review of Algorithms and Applications, 2020).
71. Khalid, R. & Javadi, N. A survey on hyperparameters optimization algorithms of forecasting models in smart grid. *Sustain. Cities Soc.* **61**, 102275. <https://doi.org/10.1016/j.scs.2020.102275> (2020).
72. Kaur, B. et al. Fostering inclusivity through effective communication: Real-time sign Language to speech conversion system for the deaf and hard-of-hearing community. *Multimed Tools Appl.* <https://doi.org/10.1007/s11042-023-17372-9> (2023).
73. Shaheen, S. Global nuclear (Back-Up) hotline: Building communication norms across NC2. *J. Peace Nucl. Disarm.* **4**, 141–162. <https://doi.org/10.1080/25751654.2021.1918932> (2021).
74. Moric, I. Nuclear stability in a world with overhead transparency. *Comp. Strateg.* **42**, 621–654. <https://doi.org/10.1080/01495933.2023.2236489> (2023).
75. Johnson, J. Deterrence in the age of artificial intelligence & autonomy: a paradigm shift in nuclear deterrence theory and practice? *Def. Secur. Anal.* **36**, 422–448. <https://doi.org/10.1080/14751798.2020.1857911> (2020).
76. Gholami Doborjeh, Z., Kasabov, N., Gholami Doborjeh, M. & Sumich, A. Modelling Peri-Perceptual brain processes in a deep learning spiking neural network architecture. *Sci. Rep.* **8**, 8912. <https://doi.org/10.1038/s41598-018-27169-8> (2018).
77. Houssein, E. H., Mohamed, R. E. & Ali, A. A. Machine learning techniques for biomedical natural Language processing: A comprehensive review. *IEEE Access.* **9**, 140628–140653. <https://doi.org/10.1109/ACCESS.2021.3119621> (2021).
78. Davies, M. et al. Advancing neuromorphic computing with loihi: A survey of results and outlook. *Proc. IEEE.* **109**, 911–934. <https://doi.org/10.1109/JPROC.2021.3067593> (2021).
79. Han, Y. et al. Dynamic neural networks: A survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **44**, 7436–7456. <https://doi.org/10.1109/TPAMI.2021.3117837> (2022).
80. Gebicke-Haerter, P. J. The computational power of the human brain. *Front. Cell. Neurosci.* **17** <https://doi.org/10.3389/fncel.2023.1220030> (2023).
81. Shen, C. A transdisciplinary review of deep learning research and its relevance for water resources scientists. *Water Resour. Res.* **54**, 8558–8593. <https://doi.org/10.1029/2018WR022643> (2018).
82. He, W. et al. Comparing SNNs and RNNs on neuromorphic vision datasets: similarities and differences. *Neural Netw.* **132**, 108–120. <https://doi.org/10.1016/j.neunet.2020.08.001> (2020).
83. Lehman, J., Clune, J. & Risi, S. An anarchy of methods: current trends in how intelligence is abstracted in AI. *IEEE Intell. Syst.* **29**, 56–62. <https://doi.org/10.1109/MIS.2014.92> (2014).
84. Doerig, A. et al. The neuroconnectionist research programme. *Nat. Rev. Neurosci.* **24**, 431–450. <https://doi.org/10.1038/s41583-023-00705-w> (2023).
85. Hussain, L. et al. Deep convolutional neural networks accurately predict breast cancer using mammograms. *Waves Random Complex. Media.* 1–24. <https://doi.org/10.1080/17455030.2023.2189485> (2023).
86. Iqbal, S. et al. Prostate Cancer detection using deep learning and traditional techniques. *IEEE Access.* **9**, 27085–27100. <https://doi.org/10.1109/ACCESS.2021.3057654> (2021).
87. Mahmood Khan, R. N. et al. COVID-19 lung infection detection using deep learning with transfer learning and ResNet101 features extraction and selection. *Waves Random Complex. Media.* 1–24. <https://doi.org/10.1080/17455030.2022.2091807> (2022).
88. Dong, R., Li, X., Zhang, Q. & Yuan, H. Network intrusion detection model based on multivariate correlation analysis – long short-time memory network. *IET Inf. Secur.* **14**, 166–174. <https://doi.org/10.1049/iet-ifs.2019.0294> (2020).
89. Yin, C., Zhu, Y., Fei, J. & He, X. A deep learning approach for intrusion detection using recurrent neural networks. *IEEE Access.* **5**, 21954–21961. <https://doi.org/10.1109/ACCESS.2017.2762418> (2017).
90. Kasongo, S. M. & Sun, Y. A deep learning method with wrapper based feature extraction for wireless intrusion detection system. *Comput. Secur.* **92**, 101752. <https://doi.org/10.1016/j.cose.2020.101752> (2020).
91. Hnamte, V., Nhung-Nguyen, H., Hussain, J. & Hwa-Kim, Y. A novel Two-Stage deep learning model for network intrusion detection: LSTM-AE. *IEEE Access.* **11**, 37131–37148. <https://doi.org/10.1109/ACCESS.2023.3266979> (2023).
92. Kim, T. & Pak, W. Early detection of network intrusions using a GAN-Based One-Class classifier. *IEEE Access.* **10**, 119357–119367. <https://doi.org/10.1109/ACCESS.2022.3221400> (2022).
93. Ahmad, Z. et al. Network intrusion detection system: A systematic study of machine learning and deep learning approaches. *Trans. Emerg. Telecommun Technol.* **32** <https://doi.org/10.1002/ett.4150> (2021).
94. Kasongo, S. M. A deep learning technique for intrusion detection system using a recurrent neural networks based framework. *Comput. Commun.* **199**, 113–125. <https://doi.org/10.1016/j.comcom.2022.12.010> (2023).
95. Kasongo, S. M. & Sun, Y. A deep learning method with filter based feature engineering for wireless intrusion detection system. *IEEE Access.* **7**, 38597–38607. <https://doi.org/10.1109/ACCESS.2019.2905633> (2019).
96. Li, Z. & Qin, Z. A Semantic Parsing Based LSTM Model for Intrusion Detection. pp 600–609 (2018).
97. Li, Z., Rios, A. L. G., Xu, G. & Trajkovic, L. Machine Learning Techniques for Classifying Network Anomalies and Intrusions. In: 2019 IEEE International Symposium on Circuits and Systems (ISCAS). IEEE, pp 1–5 (2019).

98. Wang, S., Xia, C. & Wang, T. A Novel Intrusion Detector Based on Deep Learning Hybrid Methods. In: 2019 IEEE 5th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing, (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS). IEEE, pp 300–305 (2019).
99. Tama, B. A. & Rhee, K.-H. An in-depth experimental study of anomaly detection using gradient boosted machine. *Neural Comput. Appl.* **31**, 955–965. <https://doi.org/10.1007/s00521-017-3128-z> (2019).
100. Hsu, C.-M. et al. Using Long-Short-Term memory based convolutional neural networks for network intrusion detection. In *Proceedings 2019 11th Wireless Internet: 11th EAI International Conference, WiCON 2018* 86–94 (Springer International Publishing, Taipei, Taiwan, 2018).
101. Miao, S., Wang, J. Z. & Liao, R. Convolutional neural networks for robust and Real-Time 2-D/3-D registration. In *Deep Learning for Medical Image Analysis* 271–296 (Academic Press, 2017).
102. Tang, T. A. et al. Deep learning approach for Network Intrusion Detection in Software Defined Networking. In: 2016 International Conference on Wireless Networks and (WINCOM). IEEE, pp 258–263 (2016).
103. Hosseini, S. A new machine learning method consisting of GA-LR and ANN for attack detection. *Wirel. Networks.* **26**, 4149–4162. <https://doi.org/10.1007/s11276-020-02321-3> (2020).
104. Su, T. et al. BAT: deep learning methods on network intrusion detection using NSL-KDD dataset. *IEEE Access.* **8**, 29575–29585. <https://doi.org/10.1109/ACCESS.2020.2972627> (2020).
105. Gao, X. et al. An adaptive ensemble machine learning model for intrusion detection. *IEEE Access.* **7**, 82512–82521. <https://doi.org/10.1109/ACCESS.2019.2923640> (2019).
106. Hu, Z. et al. A novel wireless network intrusion detection method based on adaptive synthetic sampling and an improved convolutional neural network. *IEEE Access.* **8**, 195741–195751. <https://doi.org/10.1109/ACCESS.2020.3034015> (2020).
107. Tama, B. A., Comuzzi, M. & Rhee, K.-H. TSE-IDS: A Two-Stage classifier ensemble for intelligent Anomaly-Based intrusion detection system. *IEEE Access.* **7**, 94497–94507. <https://doi.org/10.1109/ACCESS.2019.2928048> (2019).
108. Bostani, H. & Sheikhan, M. Hybrid of binary gravitational search algorithm and mutual information for feature selection in intrusion detection systems. *Soft Comput.* **21**, 2307–2324. <https://doi.org/10.1007/s00500-015-1942-8> (2017).
109. Almasoudy, F. H., Al-Yaseen, W. L. & Idrees, A. K. Differential evolution wrapper feature selection for intrusion detection system. *Procedia Comput. Sci.* **167**, 1230–1239. <https://doi.org/10.1016/j.procs.2020.03.438> (2020).
110. Sarumi, O. A., Adetunmbi, A. O. & Adetoye, F. A. Discovering computer networks intrusion using data analytics and machine intelligence. *Sci. Afr.* **9**, e00500. <https://doi.org/10.1016/j.sciaf.2020.e00500> (2020).

Acknowledgements

The authors extend their appreciation to the Deputyship for Research & Innovation, Ministry of Education in Saudi Arabia for funding this research work through the project number MoE-IF-UJ-R2-22-04100708-1.

Author contributions

F.S.A. Wrote the manuscript, analysis, design, implementation, visualisation, editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to F.S.A.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025